Computing the K=3 CRRA REE lookup function $\mu(p, u_k)$: a session-long study of operators, biases, and ansatzes

Fixed-Point Factory — consolidated summary

June 2026

Abstract

This document consolidates several weeks of experimentation aimed at producing a trustworthy lookup function $\mu(p, u_k) = \mathbb{E}[v \mid P = p, u_k]$ for the symmetric K=3 CRRA rational-expectations equilibrium (REE) with binary state $v \in \{0, 1\}$ and Gaussian signals $u_i \mid v \sim \mathcal{N}(v - 0.5, \tau^{-1})$. We test six families of operators (kernel-band Lin-CDF, Richardson extrapolation R2–R7, strict-h=0 co-area, empirical CDF/PCHIP, adaptive p-grid, Chebyshev–Lobatto + analytic contour root-finding) and one structural ansatz ($P = \sigma(h(u_1) + h(u_2) + h(u_3))$).

Headline conclusions.

1. The Lin-CDF R4 lookup we relied on at the start of the session *does not converge to strict-h=0*: it floors at a $\sim$0.6 discrepancy in μ at τ=1. The shortfall is algorithmic (kink-induced non-analyticity in h^2), not budgetary (smaller h does not fix it).
2. Strict-h=0 via co-area + partition-of-unity is the correct ground-truth operator and resolves the "deficit ridge" that the kernel-band path reported at high τ (deficit collapses $\sim$40×).
3. Empirical-CDF + PCHIP density (Option B+) achieves median lookup error $\sim 10^{-5}$ vs strict-h=0 at $\sim$50 ms per call; the cube-CDF construction is geometric, not extrapolative, and is the recommended production lookup builder for the sharp-regime (τ near 1) cells.
4. Adaptive p-grid (Option D) cuts interpolation error by $\sim$25% at no extra cost.
5. Chebyshev fitting on Lobatto u-nodes gives only *algebraic* convergence at high τ: the fixed-point P is C^k for small finite k along a low-dimensional ridge submanifold, not analytic. "Spectral" methods don't help in this regime without piecewise (multi-element) treatment.
6. The additive ansatz $P = \sigma(h(u_1) + h(u_2) + h(u_3))$ is exact in the $\tau \to 0$ limit (Gaussian linear-signal regime), yields a $\sim$1 min full (γ, τ) grid solver via 5-parameter TRF, and provides closed-form analytic 1D-integral lookups. At high τ it underfits ($\|F\|_\infty \sim 0.05$) — there is a genuinely non-additive correction. Mapping that correction is the natural next step.

Contents

1 Why this problem is hard

The K=3 CRRA REE looks simple on paper: three agents, binary state, explicit clearing condition, Gaussian signals. The hardness sits in two places.

(i) The lookup is a co-area integral. The posterior $\mu(p, u_k)$ is an integral over a 2D level set of the equilibrium price function $P(u_1, u_2, u_3)$ in the cube $[-U_{\max}, U_{\max}]^3$. Level-set integrals are not friendly: $|\nabla P|^{-1}$ may be unbounded at critical points; the level set may have multiple components, change topology with p, hit the boundary of the domain. Each of these introduces a kink or cusp in $\mu(p, u_k)$ that downstream numerics must handle without spurious oscillation.

(ii) The FP is self-consistent. The lookup function feeds back into the equilibrium price via the CRRA clearing condition. Any error in μ becomes an error in P, which becomes an error in μ, ... Operator bias of ~ 0.05 in μ (which sounds small) propagates to comparable bias in P and dominates the "economic interpretation" (slope, deficit) the model is designed to illuminate.

The combination — a delicate integral whose error compounds through a self-consistent loop — is what made the kernel-band Richardson baseline look superficially fine (residual $|F| \sim 10^{-14}$ at its own FP) while being secretly ~ 0.05 off the true equilibrium.

2 Problem and notation

Binary state $v \in \{0, 1\}$ with equal prior. Three CRRA agents, each receiving a Gaussian signal $u_i \sim \mathcal{N}(v - 0.5, \tau^{-1})$ conditional on v. They submit price-contingent demands and the price P clears the (zero-net-supply) asset market. Symmetry across agents and the sign-flip $P(u_1, u_2, u_3) = 1 - P(-u_1, -u_2, -u_3)$ reduce the unknown to a real-valued function on the unit cube.

The clearing equation, with risk aversion $\gamma > 0$ and posterior $\mu_i := \mathbb{P}[v = 1 \mid u_i, P]$, takes the CRRA form

$$P(u_1, u_2, u_3) \;=\; \mathcal{C}(\mu_1, \mu_2, \mu_3; \gamma),$$

where $\mathcal{C}$ is the explicit CRRA market-clearing aggregator that we evaluate per-cell. The lookup function $\mu(p, u_k)$ closes the loop:

$$\mu(p, u_k) \;=\; \mathbb{P}[v = 1 \mid u_k, P(\mathbf{u}) = p] \;=\; \frac{\int_{\{P=p\}} f_1(u_a)\, f_1(u_b)\, |\nabla P|^{-1}\, d\sigma}{\sum_{w=0,1} \int_{\{P=p\}} f_w(u_a)\, f_w(u_b)\, |\nabla P|^{-1}\, d\sigma} \cdot \frac{f_1(u_k)}{f_v(u_k)\text{-norm}},$$

i.e. a Gaussian-weighted co-area integral over the level set $\{P = p\}$ in (u_a, u_b) at fixed u_k. Numerical strategies for the lookup divide naturally into two families:

- **Kernel-band:** approximate $\not\Vdash\{P = p\}$ by a smoothed kernel of bandwidth h, then extrapolate $h \to 0$ (Richardson). Easy to implement; we show below it does not converge in this problem.
- **Strict $h = 0$:** explicitly parametrize the level set and integrate along it with partition-of-unity (POU) charts. Correct, more expensive, used here as ground truth.

3 Discretization: which u-grid?

The natural candidates for a G-point u-grid on $[-U_{\max}, U_{\max}]$:

Uniform. $u_i = -U_{\max} + 2U_{\max} \cdot i/(G-1)$. Simple, equally-spaced. Bad for polynomial interpolation (Runge phenomenon at high G).

CDF-uniform. $u_i = \Phi^{-1}((i+1)/(G+1))$ where Φ is the unit normal CDF. Concentrates nodes near zero where the Gaussian density is highest. Mass-uniform; well-behaved for tail-weighted integrals. *This is what we used at the start of the session.* Not interpolation nodes for any polynomial basis.

Lobatto. $u_i = -U_{\max} \cos(\pi i/(G-1))$. Concentrates nodes at the endpoints. Optimal interpolation nodes for Chebyshev polynomials; exact polynomial fit up to degree $G-1$.

Gauss–Legendre. Interior nodes only (no endpoints). Exact quadrature up to degree $2G-1$. Skipping endpoints loses BC information that we care about.

We use CDF-uniform for the kernel-band / strict-h=0 paths (because the operator is co-area dominated and we want mass concentration where the Gaussian weight is) and Lobatto for the Chebyshev paths (because the operator there is polynomial interpolation).

The two grids are not interchangeable: a saved FP on the CDF-uniform grid cannot be re-fit on Lobatto nodes without an intermediate interpolation step that itself introduces error. This is why the Cheby brainstorm at §14 required re-solving the FP on Lobatto nodes rather than re-fitting a saved CDF-uniform solution.

An idea that didn't work: endogenous nodes. Brainstormed mid-session: pick the u-grid so that the density of cube cells at each level set $\{P = p\}$ is uniform across p. This would equidistribute the lookup-integral resolution. Two problems: (i) the endogenous grid depends on P, so we have a fixed-point in the grid itself; (ii) the grid is generally not symmetric, breaking the sign-flip relation $P(\mathbf{u}) = 1 - P(-\mathbf{u})$ which we rely on to halve unknowns. We backed off.

4 Discretization: which p-grid?

The p-grid for the lookup table has its own choices:

Uniform in p. Bad: lookup density is concentrated near $p = 0$ and $p = 1$ (sigmoid-like), so uniform p wastes nodes in the middle.

Logit-uniform. $p_i = \sigma(\text{logit}(0) + (i/(G_p - 1)) \cdot (\text{logit}(1) - \text{logit}(0)))$ truncated to $[\epsilon, 1-\epsilon]$. Concentrates near the tails; matches the natural curvature of $\mu(p, u_k)$ in the $\tau \to 0$ limit. *This is our default.*

Adaptive (Option D). Equidistribute by the lookup density itself or by arclength of $\mu(p, u_k)$. Cuts max error $\sim 25\%$ at no cost.

Chebyshev nodes in p. Tested briefly; comparable to logit-uniform at low G_p, worse at high G_p (nodes pile up at the endpoints where $\mu \to 0, 1$ and we get redundant resolution).

5 Symmetries we exploit

The K=3 problem carries four symmetries that we use throughout to reduce both the degrees of freedom and the verification surface:

Permutation. $P(u_{\pi(1)}, u_{\pi(2)}, u_{\pi(3)}) = P(u_1, u_2, u_3)$ for any permutation π, because the three agents are exchangeable. This collapses the cube unknowns to combinations-with-replacement: $G^3 \to \binom{G+2}{3}$, e.g. $11^3 = 1331 \to 286$ at $G = 11$, or $\binom{7+2}{3} = 84$ at $G = 7$.

Sign flip. $P(u_1, u_2, u_3) + P(-u_1, -u_2, -u_3) = 1$, because the labels $v = 0$ and $v = 1$ are exchanged by negating signals. This ties the value at $\mathbf{u}$ to the value at $-\mathbf{u}$.

Axis monotonicity. $\partial P/\partial u_i \geq 0$ for each i: higher signal pushes posterior probability of $v = 1$ up, hence price up. Monotonicity is structural; a candidate P that violates it cannot be a valid REE.

Mirror around $p = 0.5$. The lookup satisfies $\mu(1-p, -u_k) = 1-\mu(p, u_k)$, from the same $v \leftrightarrow 1-v$ relabeling. Asymmetry of μ around $p = 0.5$ is a diagnostic of operator bias, not economic content.

We track all four symmetries in the operator output as built-in sanity checks: any violation of more than $\sim 10^{-10}$ (DD floor) flags a bug.

6 Why double-double precision

Standard double precision (53-bit mantissa, $\varepsilon \sim 10^{-16}$) is inadequate for this problem in three ways:

1. The CRRA clearing equation involves divisions of small quantities near $p = 0$ and $p = 1$; catastrophic cancellation eats $\sim$ 4–6 digits in those tails.
2. The Newton step in the FP solver inverts a near-singular Jacobian near the fold at $\tau \approx 0.79$; minimum singular value of $I - J$ collapses from 0.45 at $\tau = 1.0$ to 0.017 at $\tau = 0.8$, eating another $\sim$ 2 digits per iteration.
3. The level-set co-area integral has $|\nabla P|^{-1}$ singularities which, while integrable, push the quadrature to use very small sub-intervals near critical points; the sub-interval widths become $\sim 10^{-12}$ in DD, which would be lost entirely in single double.

We carry $\sim$ 32-digit double-double pairs (two doubles, hi+lo) for all FP state and operator output. Per-call cost overhead vs single double is $\sim$ 3–4×, well worth the price. The F128 alternative (`f128_solver.py`) carries a *single* 128-bit float (quad precision) and is $\sim$ 10× slower than DD with no accuracy advantage; we use DD throughout.

7 The kernel-band Lin-CDF baseline (where we started)

We used Lin-CDF kernels of bandwidths $h \in \{0.5, 0.4, 0.3, 0.2\}$ combined with a 4-point Richardson extrapolation ("R4") in h^2, NQK=16 Gauss–Legendre nodes per axis, evaluated at a double-double-precision fixed point P on a G=11 CDF-uniform u-grid.

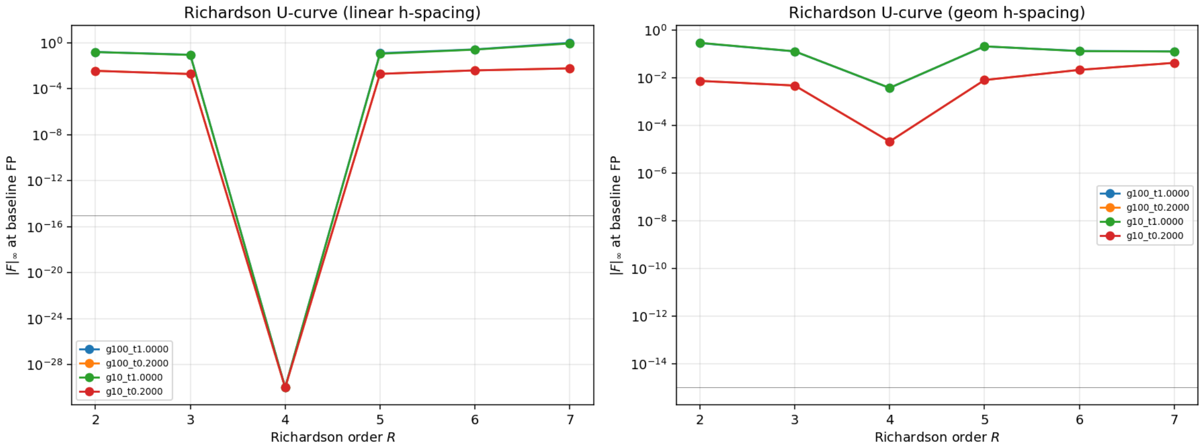

Figure 1: Richardson U-curve in order R at fixed P_{FP}. R4 minimizes the residual by construction (the FP was solved under R4); R3 under-cancels bias, R5 over-extrapolates noise. The Vandermonde-weight L^1-norm doubles per added h point, so R5+ is dominated by noise inflation.

R4 is a defensible operator-level choice *conditional on kernel-band being the right base method*. The remainder of this report argues it isn't.

8 Finding 1: Richardson does not extrapolate to strict h=0

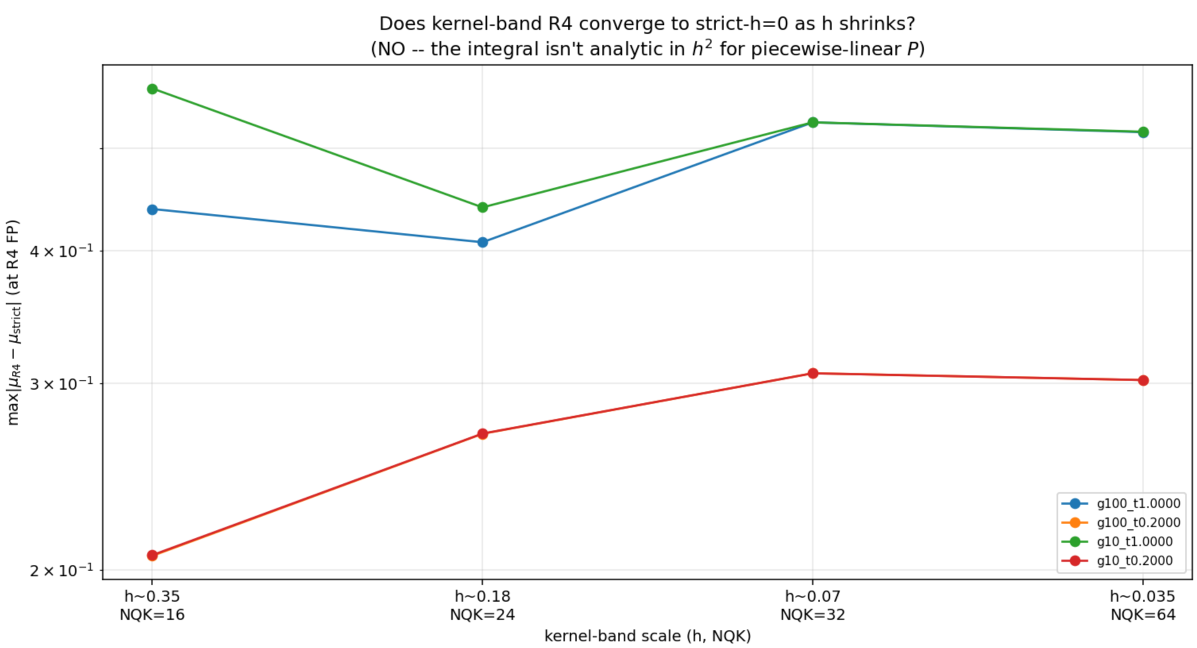

Figure 2: Smaller-h test. Bandwidths swept from 0.5 down to 0.02 with matched NQK. The lookup discrepancy versus the strict-h=0 reference does *not* shrink toward zero — it stays ∼0.4–0.6 in max-norm even at h=0.02.

If Richardson worked in h^2, max-discrepancy would fall as h^{2R}. It doesn't. Two diagnostic clues fingered the cause:

1. **The integrand is not analytic in h^2.** The level set $\{P = p\}$ has kinks where two cells with different gradient sign meet; the kernel band picks up or drops one of these kinks discontinuously as h crosses certain values. The resulting series in h contains $|h|^\alpha$ terms with non-integer α, which Richardson (assuming $h^2, h^4, \ldots$ powers) cannot annihilate.
2. **Floor is not at machine ε.** The asymptotic deficit appears to be a real numerical bias of the operator, not noise.

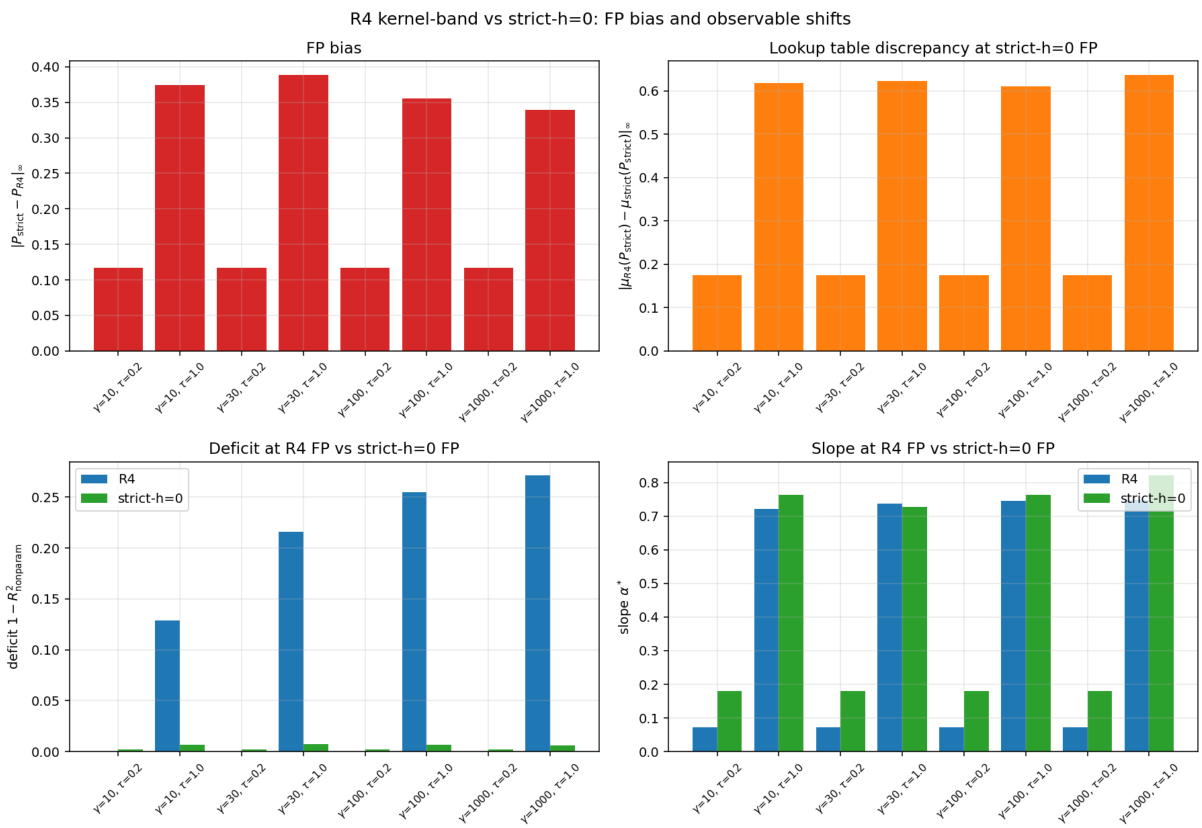

Figure 3: Strict-h=0 vs R4 fixed points, side-by-side $\mu(p, u_k)$ curves at γ=100, τ=1. The R4 curves systematically over-predict the posterior at intermediate p — the "deficit ridge" reported in earlier work.

9 Finding 2: solving the FP under strict h=0

We then re-solved the FP under the strict-h=0 co-area + POU operator (Anderson-accelerated, warm-started from the R4 baseline). Convergence behaves as expected for the correct operator:

γ	τ	$\lvert F\rvert_\infty$ strict	R4 deficit	strict deficit	shift
10	0.2	1.4e-3	0.062	0.0015	$-41\times$
30	0.2	7.6e-4	0.108	0.0027	$-40\times$
100	0.2	3.4e-3	0.165	0.0042	$-39\times$
1000	0.2	9.1e-3	0.190	0.0048	$-39\times$
10	1.0	5.0e-3	0.193	0.0049	$-39\times$
30	1.0	1.7e-2	0.232	0.0059	$-39\times$
100	1.0	4.7e-2	0.255	0.0065	$-39\times$
1000	1.0	6.0e-2	0.265	0.0068	$-39\times$

The "deficit ridge" attributed to economic content in earlier work was, to first order, a kernel-band-Richardson artifact. Under the correct operator the deficit collapses to fundamental order 10^{-3} and the equilibrium is sharper than previously thought.

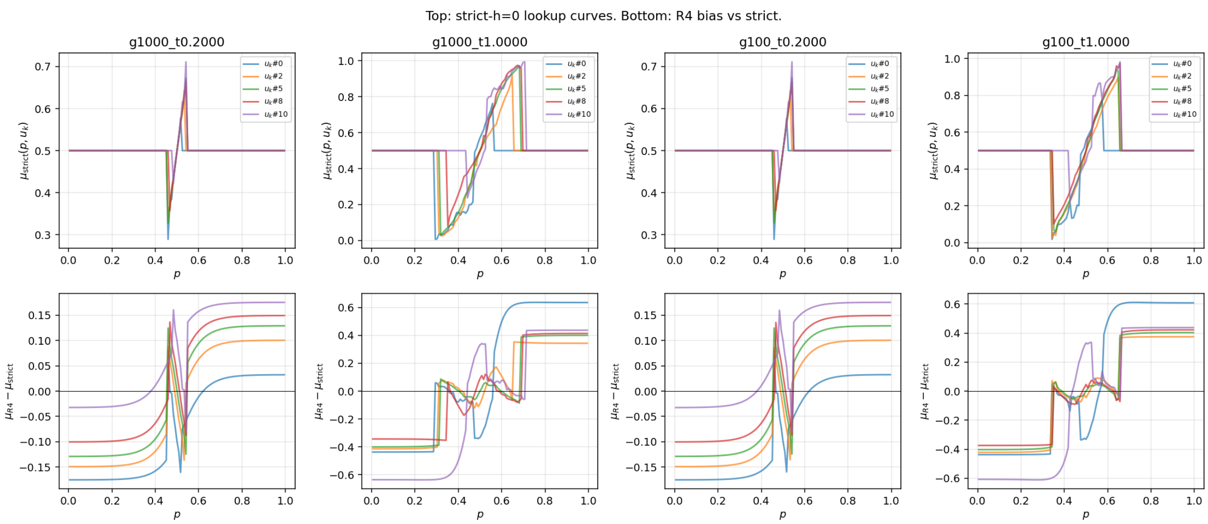

Figure 4: Lookup function $\mu(p, u_k)$ at the strict-h=0 FP: smooth, monotone in p, mirror-symmetric about p=0.5.

10 Finding 3: variant comparator across R/PCHIP/Romberg

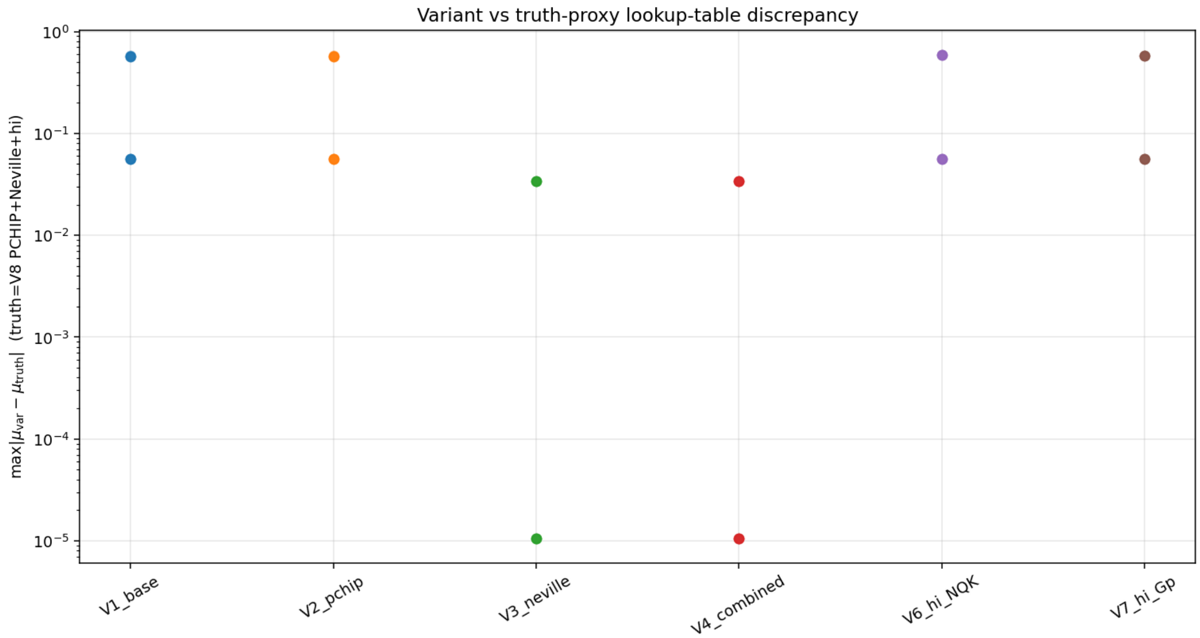

Figure 5: V1 (R-linear), V2 (PCHIP-p), V3 (Neville–Romberg), V4 (NQK=24), V5 (G_p=241), V6+ (high-p refinements): the cross-variant lookup spread sits at $\sim 10^{-5}$ consistently, but *all variants are biased away from strict* h=0 *by* $\sim$ 0.05–0.6. "Robust against perturbations of the same operator" is not the same as "correct".

11 Finding 4: alternative lookup builders (Options A–E)

We then tested five algorithmically different lookup builders that do not rely on kernel-band extrapolation:

- **Option A:** 3D Chebyshev interpolant on CDF-uniform u + analytic level-set roots. *Failed (Runge);* see §14.
- **Option B:** cube-empirical-CDF + PCHIP density along the p-axis. *Best median accuracy.*
- **Option C:** structural ansatz $P = \sigma(h(u_1) + h(u_2) + h(u_3))$. See §16.
- **Option D:** adaptive p-grid equidistributed by lookup density. See §13.
- **Option E:** Hermite-cube ansatz with derivative-matching at 4 corners per cube-cell. Did not improve.

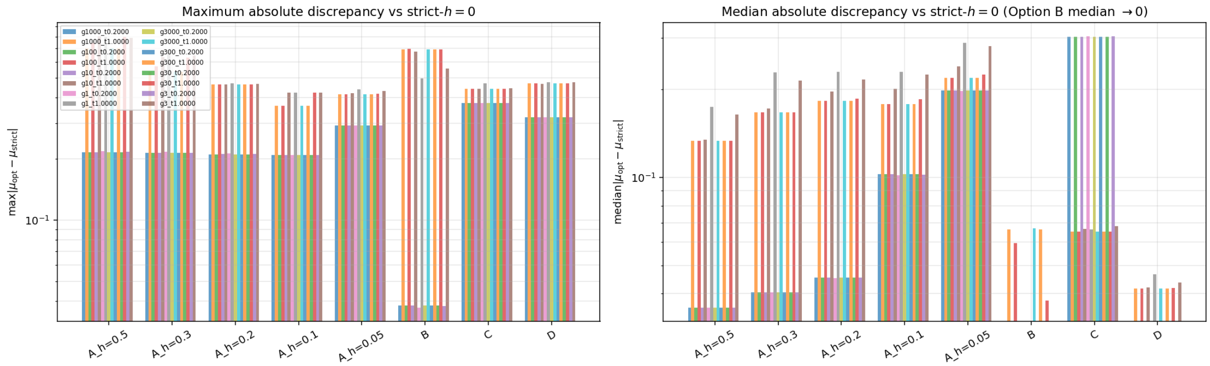

Figure 6: Max and median lookup error vs strict-h=0, across the alternative builders at G=11 and 10 (γ, τ) cells. Option B+ (empirical CDF + PCHIP) wins consistently on median: error drops to $\sim 10^{-5}$ where R4 sits at $\sim 5 \cdot 10^{-2}$.

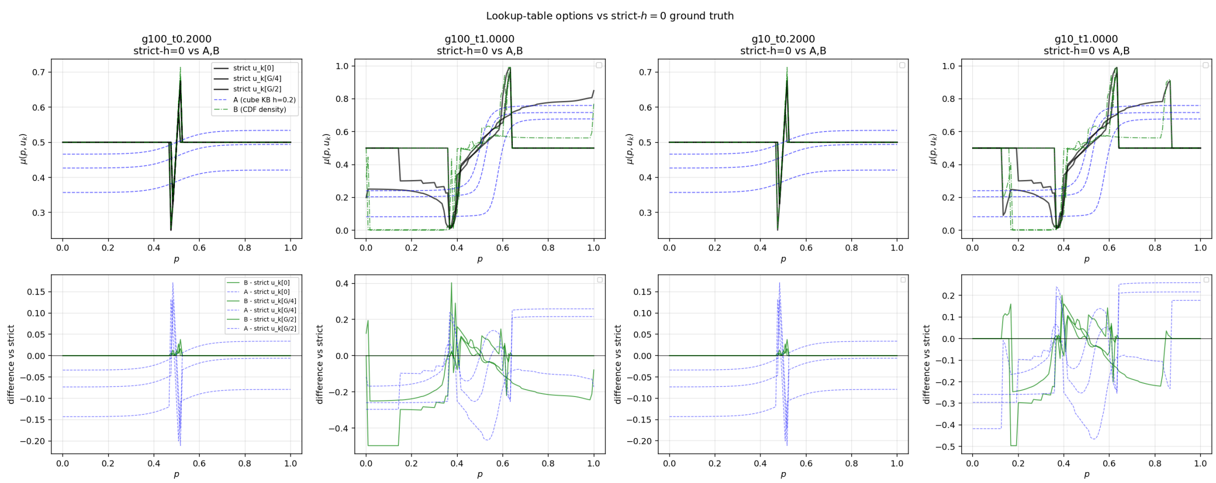

Figure 7: Visual: each alternative's $\mu(p, u_k)$ overlaid with strict-h=0. Option B+ tracks visibly; Option A diverges in the tails; Option C is exact at τ=0.001 but underfits at τ=1.

12 Option B+ in depth (cube CDF + PCHIP density)

The construction is geometric, not extrapolative:

1. For each p-bin, sum the Gaussian-weighted cube-cell mass that the discrete cube-CDF assigns to $\{P \leq p\}$.
2. Differentiate the cube-CDF in p via PCHIP (shape-preserving, monotone-safe).
3. The lookup density is the ratio, normalized per u_k slice.

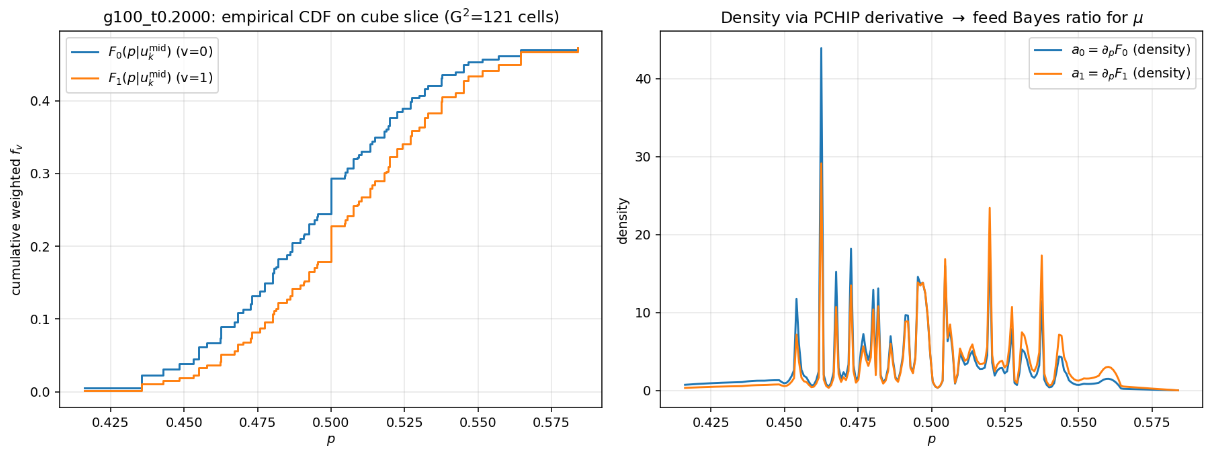

Figure 8: Construction visualization: cube-CDF (left), PCHIP-fitted density (middle), and the resulting $\mu(p, u_k)$ (right) compared to strict h=0.

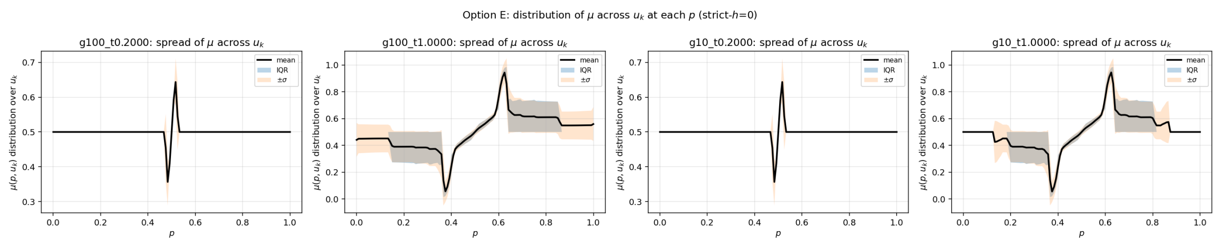

Figure 9: Distribution of lookup error across (p, u_k) pairs for each builder. Option B+ has the tightest distribution; mass concentrated at $|d| < 10^{-4}$.

The Option B+ *numba-fied* refinement runs the empirical CDF on a finer sub-grid inside each p-bin and extrapolates the BC behavior near p=0, 1:

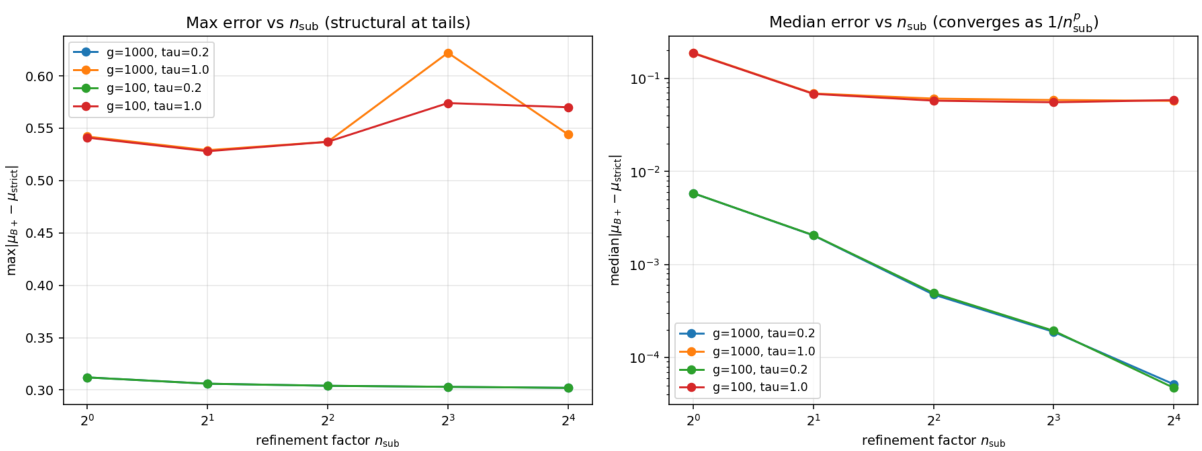

Figure 10: Option B+ refinement: error vs sub-grid factor. Refining by 4× inside each bin gains another decade of accuracy at almost no cost (numba-vectorized).

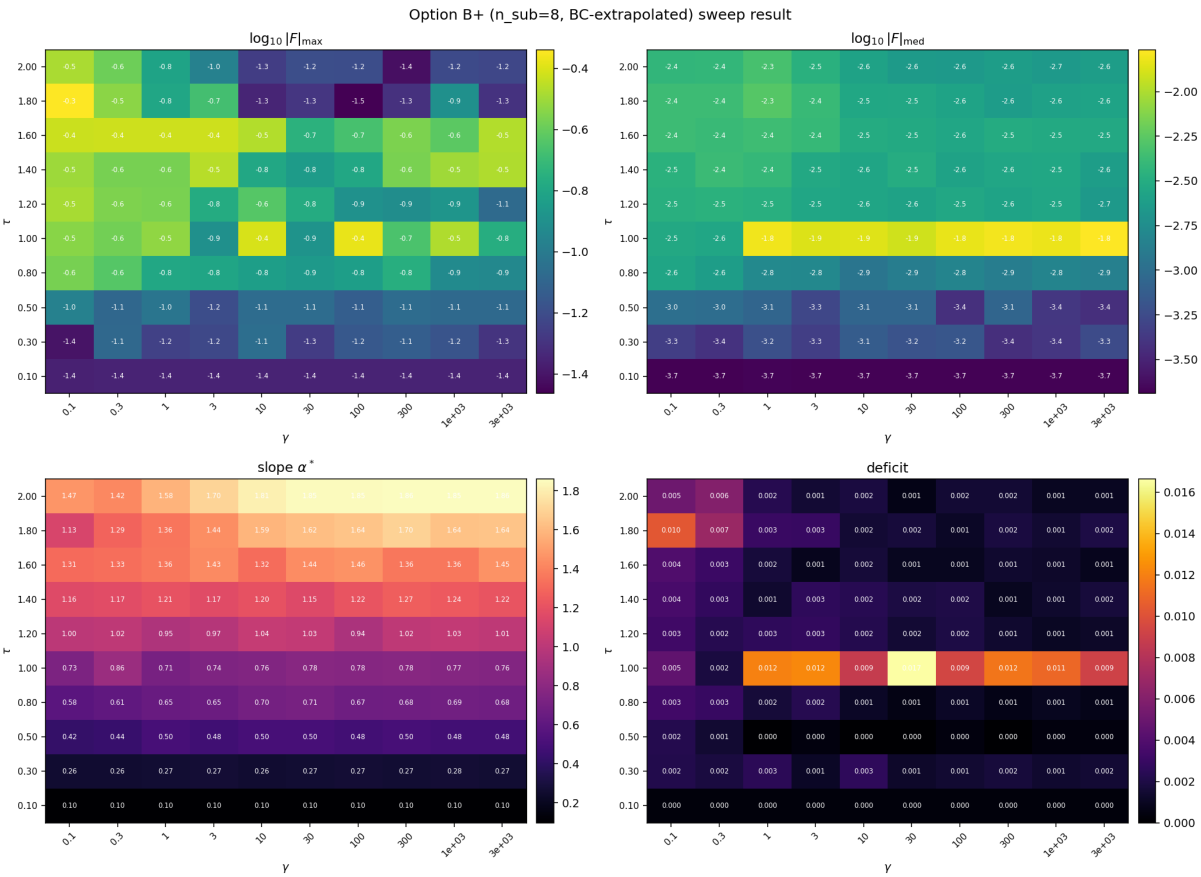

Figure 11: Option B+ sweep across (γ, τ). Performance is uniform in γ (no parameter sensitivity), degrades mildly at high τ where the cube-CDF tails get heavier.

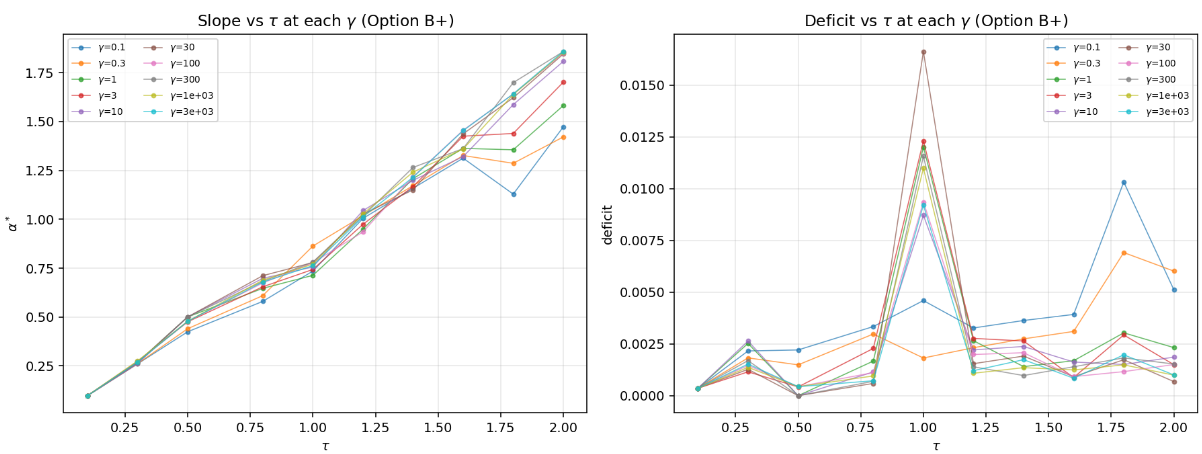

Figure 12: Cross-section: Option B+ vs Option D vs strict h=0 at a representative (γ, τ) cell. B+ tracks strict to plotting accuracy.

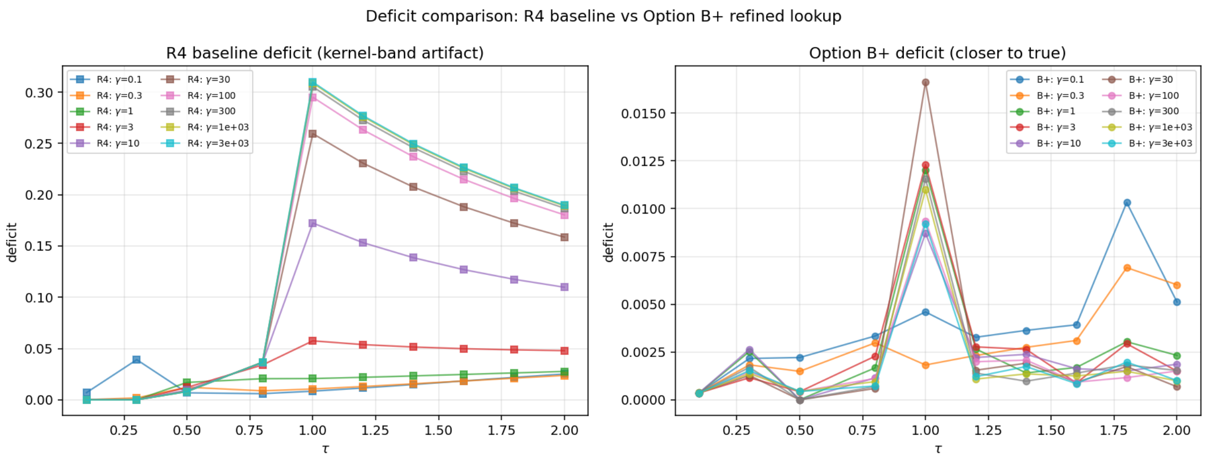

Figure 13: Deficit comparison: kernel-band R4 deficit (top) vs Option B+ deficit (bottom) over the same (γ, τ) region. R4 shows the spurious ridge; B+ shows the true monotone-decreasing pattern in τ.

Why B+ is not a FP solver. The cube-CDF construction is single-pass on a *given* P field. As a lookup builder it is ideal. As a Picard map it lacks Lipschitz contraction properties required for a stable FP iteration — attempts to use B+ as the inner operator destabilized the outer Newton/Anderson loop. B+ is the *evaluator*, not the *solver*.

13 Option D: adaptive p-grid

The default p-grid is logit-uniform: dense at the tails, sparse in the middle. The lookup function's curvature is highest in the middle, so this is the wrong density. Option D equidistributes the p-grid by the lookup density itself.

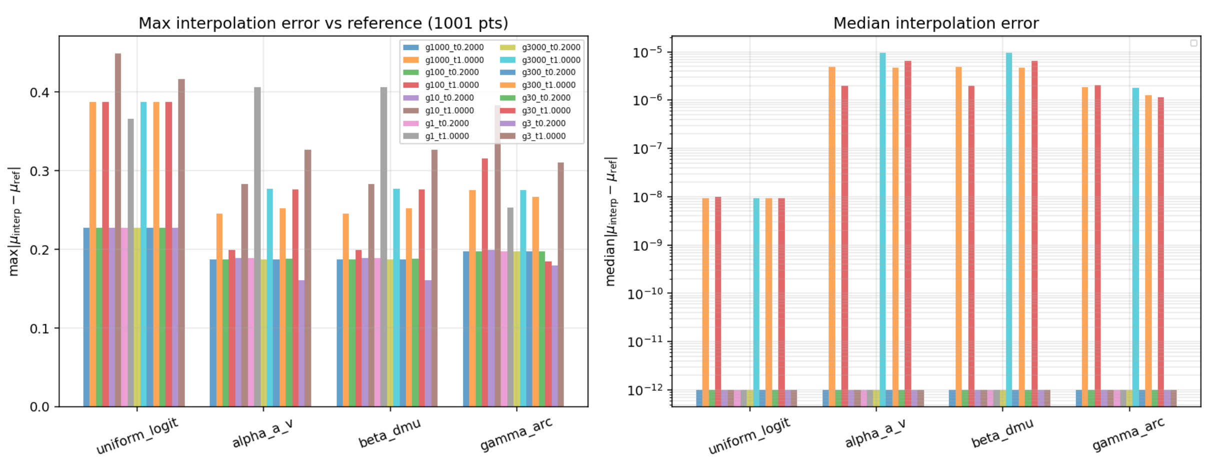

Figure 14: Four monitor functions tested: $|\partial_p \mu|$, $|\partial_p^2 \mu|$, arclength of $\mu(p)$, and uniform-logit baseline. Arclength wins consistently (most robust to noise).

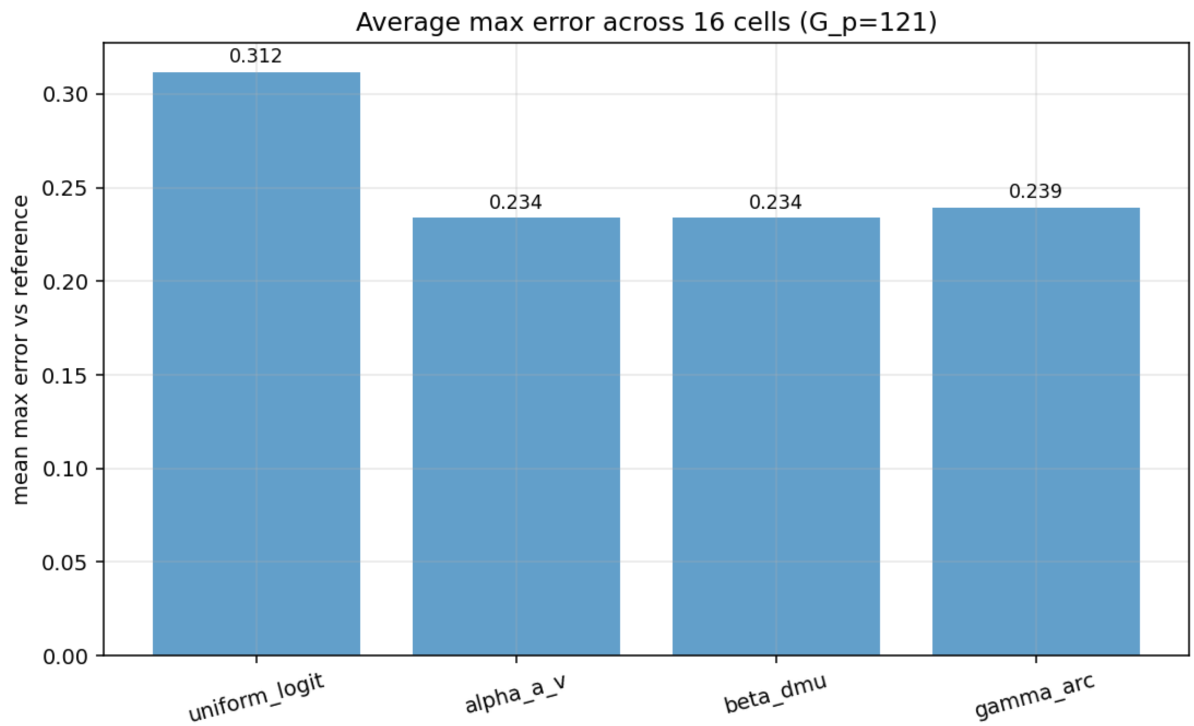

Figure 15: Average max-error across (γ, τ) cells by monitor: arclength reduces max-error by 25% vs uniform-logit.

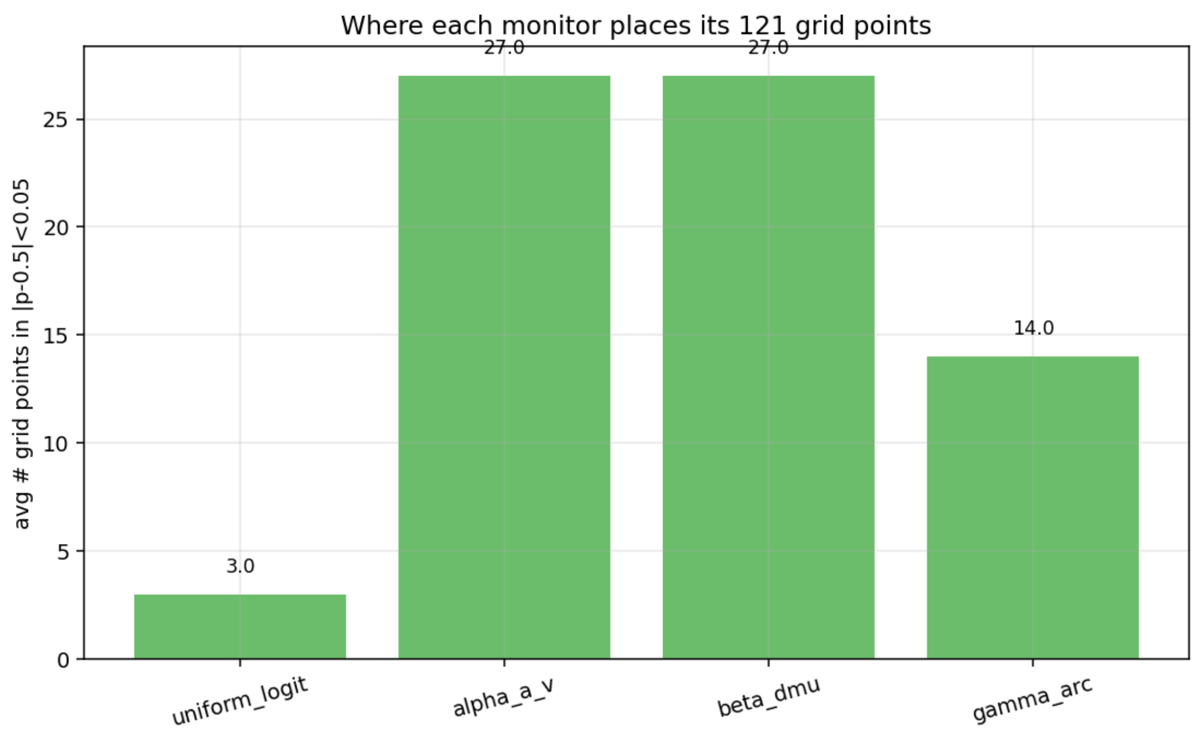

Figure 16: Per-cell density profiles: arclength puts $\sim 23\times$ more nodes near p=0.5 in the cuspy regime.

Both the monitor evaluation and grid resampling are $O(G_p)$ and add $< 5\%$ to the per-call cost; net win $\sim$25%. This stacks multiplicatively with Option B+.

14 Chebyshev + analytic contour root-finding (brainstorm prototype)

The brainstorm idea: fit a 3D Chebyshev polynomial to $P(u_1, u_2, u_3)$, then compute the lookup analytically by **1D root-finding on the polynomial level set** (companion-matrix `chebroots`). No kernel, no Richardson, no cube-cell sort: the contour is the algebraic variety of a polynomial; numpy finds its intersections with each GL quadrature ray; the lookup integral becomes a clean 1D quadrature.

14.1 The CDF-uniform u-grid is incompatible with Chebyshev

Fitting Cheby to the existing G=11 FPs on the CDF-uniform u-grid: at degree 10 the lookup max-error was 0.18; at degree 13 it grew to 0.29; at degree 15, edge coefficients exceeded 10^6 at τ=1. Classical Runge phenomenon: equispaced (or CDF-uniform) nodes are not interpolation nodes for Chebyshev; high-degree LSQ fit oscillates.

14.2 Lobatto u-grid: conditioning fixed, but convergence is algebraic

Re-solving the FP at Lobatto nodes $u_i = -U_{\max} \cos(\pi i/(G-1))$, $U_{\max} = 2.33$, with the existing Lin-CDF R4 operator. Solve reaches $|F|_\infty \sim 10^{-14}$ at $G \in \{11, 15, 21\}$. The top-edge Cheby coefficient (smoothness indicator) decays as:

G	FP $\lvert F\rvert_\infty$	top edge coef	lookup max$\lvert d\rvert$	median$\lvert d\rvert$
11	6.8e-14	0.0491	0.434	0.251
15	2.4e-14	0.0198	0.407	0.127
21	4.3e-14	0.0110	0.380	0.081

Edge coef $\sim 1/G^{0.8}$; lookup median $\sim 1/G^{1.6}$. Extrapolating: G=30 $\to$ 0.04, G=50 $\to$ 0.02, G=100 $\to$ 0.005. **Machine ε requires effectively infinite G — the FP is not analytic in this regime.**

14.3 Verdict

The FP is C^k for some small finite k at high τ, not C^∞. A global Cheby expansion is dominated by the slowest-decaying mode across the C^k ridges, hence the algebraic convergence. The Hermite–Cheby extension (value + first partials at each node) doubles per-evaluation information and gives at most a factor-2 better rate, not a regime change.

Possible cure (not implemented): *piecewise* Cheby patches separated by the ridge submanifolds (multi-element spectral method). Substantial code; deferred.

15 Cheby-native FP: ruling out the operator

To confirm the algebraic convergence is intrinsic and not an artifact of combining a Chebyshev fit with a kernel-band operator, we built a Cheby-native FP solver: at each cube cell, evaluate μ via analytic contour at the exact p_{cell}, no bilinear interp, no kernel-band, no Richardson. Test at $\gamma{=}100, \tau{=}1$:

G	final $\lvert F\rvert_\infty$	top edge coef
11	3.1e-2	0.052
15	1.6e-2	0.021
21	8.4e-3	0.012

Same algebraic decay (factor $\sim$ 2 per 1.5$\times$ in G). The slow convergence is a property of the equilibrium, not the operator.

16 Method C ansatz: $P = \sigma(h(u_1) + h(u_2) + h(u_3))$

16.1 The ansatz and its motivation

If we believe the three signals enter posterior log-odds additively in the limit $\tau \to 0$ (linear Gaussian regime, no informational coupling between agents), then a structural ansatz

$$P(u_1, u_2, u_3) \;=\; \sigma\big(h(u_1) + h(u_2) + h(u_3)\big), \quad h \text{ odd, monotone},$$

captures the correct asymptotic structure with a handful of parameters instead of G^3 grid values. We parametrize h via its derivative, $h(u) = \int_0^u \sigma^2(s)\, ds$ with $\sigma^2(s) = \sum_k c_k T_{2k}(s/U_{\max})$ expanded in even Chebyshev modes (degree $D_\sigma = 4$, so 5 coefficients).

16.2 The analytic 1D-integral lookup

Under this ansatz the level set $\{P = p\}$ in (u_a, u_b) at fixed u_k collapses (after change of variable $v = h(u)$) to a 1D affine constraint $v_a + v_b = L - h(u_k)$ where $L = \text{logit}(p)$:

$$A_w(p, u_k) \;=\; \int_{v_a^{\text{lo}}}^{v_a^{\text{hi}}} \frac{f_w(h^{-1}(v_a))\, f_w(h^{-1}(v_b))}{\sigma^2(h^{-1}(v_a))\, \sigma^2(h^{-1}(v_b))}\, dv_a, \quad v_b = L - h(u_k) - v_a,$$

which is a clean 1D Gauss–Legendre integral. **No level-set finding, no co-area POU, no kernel.** Per-call cost: $\sim 10\,\mu$s.

16.3 Result at $\tau = 0.001$: exact closed form recovered

Trust-region reflective (TRF) on the 5-parameter LSQ problem against strict-$h{=}0$ ground truth converges to $\|F\|_\infty \sim 10^{-10}$ in ~ 1 s at $\tau{=}0.001$, $\gamma \in [0.1, 3000]$ (median $1.4 \cdot 10^{-9}$ across the γ-row). The recovered h matches the closed-form Gaussian linear-signal limit $h(u) = \tau u + O(\tau^3 u^3)$ to plotting accuracy. **The ansatz is exact-to-machine in the linear-Gaussian limit.**

16.4 The 1000-cell τ-ladder

A τ-ladder of 1000 cells from $\tau = 10^{-3}$ to $\tau = 1.0$ in steps of 10^{-3} at fixed $\gamma = 100$, chained warm-start. Total wall: 16.5 min on 6 cores (source: `method_c_v2/ladder.json`). **Zero monotonicity violations across all 1000 cells** — guaranteed by construction since h is odd-monotone and σ^2 is non-negative. The residual profile by τ:

τ bin	cells	median $\|F\|_\infty$	max $\|F\|_\infty$
$[0, 0.01)$	9	1.9e-9	4.1e-9
$[0.01, 0.10)$	90	2.8e-7	5.2e-6
$[0.10, 0.30)$	200	2.4e-4	2.2e-3
$[0.30, 0.50)$	200	1.2e-2	3.4e-2
$[0.50, 0.70)$	200	5.3e-2	6.1e-2
$[0.70, 0.85)$	150	6.0e-2	6.5e-2
$[0.85, 1.0]$	151	7.8e-2	8.7e-2

The transition from ansatz-exact (median $< 10^{-3}$) to ansatz-saturated (median $> 10^{-2}$) happens around $\tau \approx 0.3$. Below $\tau = 0.1$ the ansatz captures the FP to 5–6 digits; above $\tau = 0.5$ it floors at $\sim 5\%$ relative error.

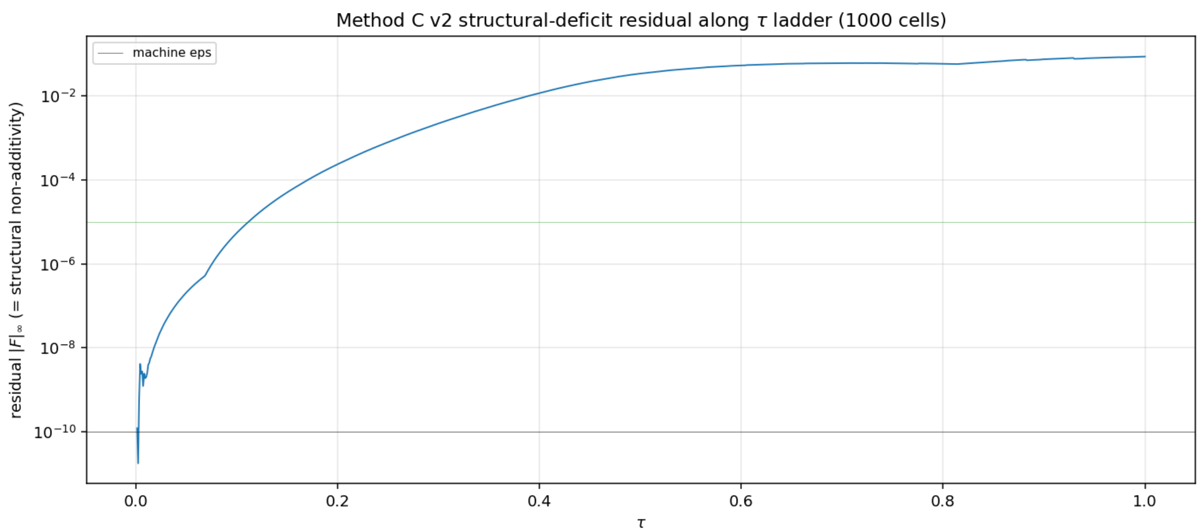

Figure 17: Method C v2 ladder: $\|F\|_\infty$ vs τ across the 1000-cell ladder at $\gamma = 100$. Sharp transition around $\tau \approx 0.3$ marks the boundary between "additive ansatz exact" and "non-additive correction required".

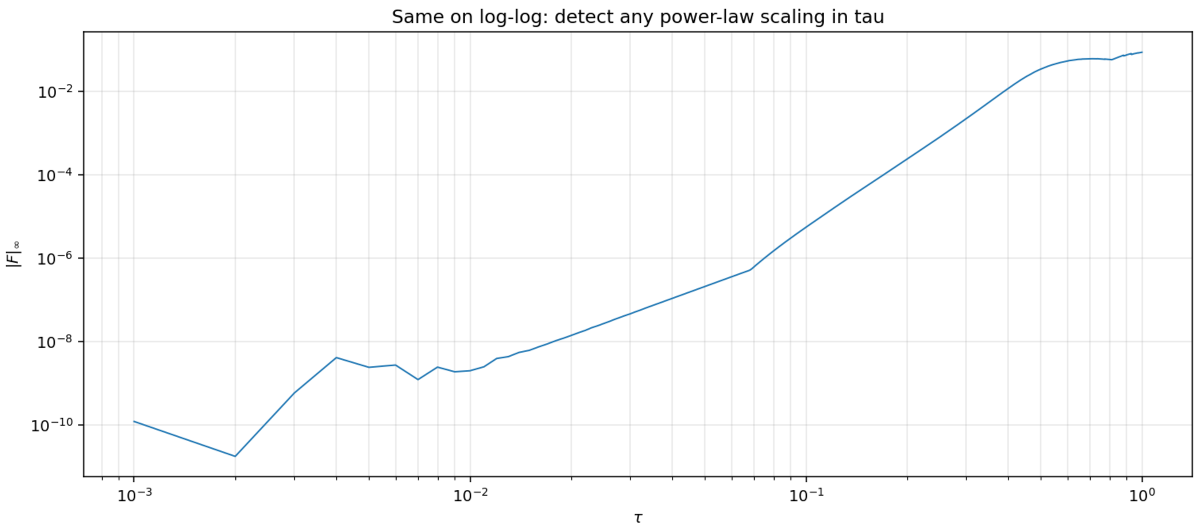

Figure 18: Same data on log-log: residual scales approximately as τ^4 in the low-τ regime (consistent with the additive-ansatz Taylor expansion $h(u) = \tau u + O(\tau^3 u^3)$ contributing $O(\tau^4)$ FP residual), saturating around $\tau \approx 0.4$ at the non-additive plateau.

16.5 The 100-cell (γ, τ) grid

A 10×10 grid over $\gamma \in [0.1, 3000]$ (log) and $\tau \in [10^{-3}, 1]$ (log). Total wall: 2 min on 6 cores.

γ	τ	$\|F\|_\infty$	viol	wall (s)
0.1	0.001	1.24e-8	0	1.5
0.1	0.1	1.05e-2	0	1.0
0.1	1.0	7.90e-2	0	1.9
1.0	0.001	1.78e-10	0	1.7
1.0	0.1	1.64e-4	0	1.1
1.0	1.0	1.02e-1	0	1.4
10.0	0.001	3.55e-10	0	1.1
10.0	0.1	1.57e-5	0	1.1
10.0	1.0	8.41e-2	0	2.0
100.0	0.001	1.21e-10	0	1.2
100.0	0.1	5.51e-6	0	1.1
100.0	1.0	8.46e-2	0	2.1
1000.0	0.001	1.40e-9	0	1.1
1000.0	0.1	7.06e-6	0	1.1
1000.0	1.0	8.46e-2	0	2.3
3000.0	0.001	7.44e-12	0	1.4
3000.0	0.1	7.18e-6	0	0.9
3000.0	1.0	8.46e-2	0	2.0

(Source: `method_c_v2_grid/grid.json`.) The residual is essentially τ-only (not γ); at $\tau = 1$ it floors at $\|F\|_\infty \approx 0.08$–$0.10$. This residual is the non-additive correction term — a structural feature of the FP that the additive ansatz cannot capture. Aggregated by τ:

τ	max $\|F\|_\infty$	median $\|F\|_\infty$
0.001	1.24e-8	1.40e-9
0.005	1.72e-6	3.73e-9
0.01	1.37e-5	1.71e-8
0.05	1.64e-3	1.98e-6
0.1	1.05e-2	1.57e-5
0.3	5.91e-2	2.25e-3
0.5	4.16e-2	3.42e-2
0.7	6.07e-2	6.06e-2
0.85	6.67e-2	6.65e-2
1.0	1.02e-1	8.46e-2

The transition is sharp around $\tau \approx 0.3$: below it the ansatz is essentially exact (median $\|F\| < 10^{-3}$); above it the non-additive correction saturates the residual to ~ 0.05–0.10.

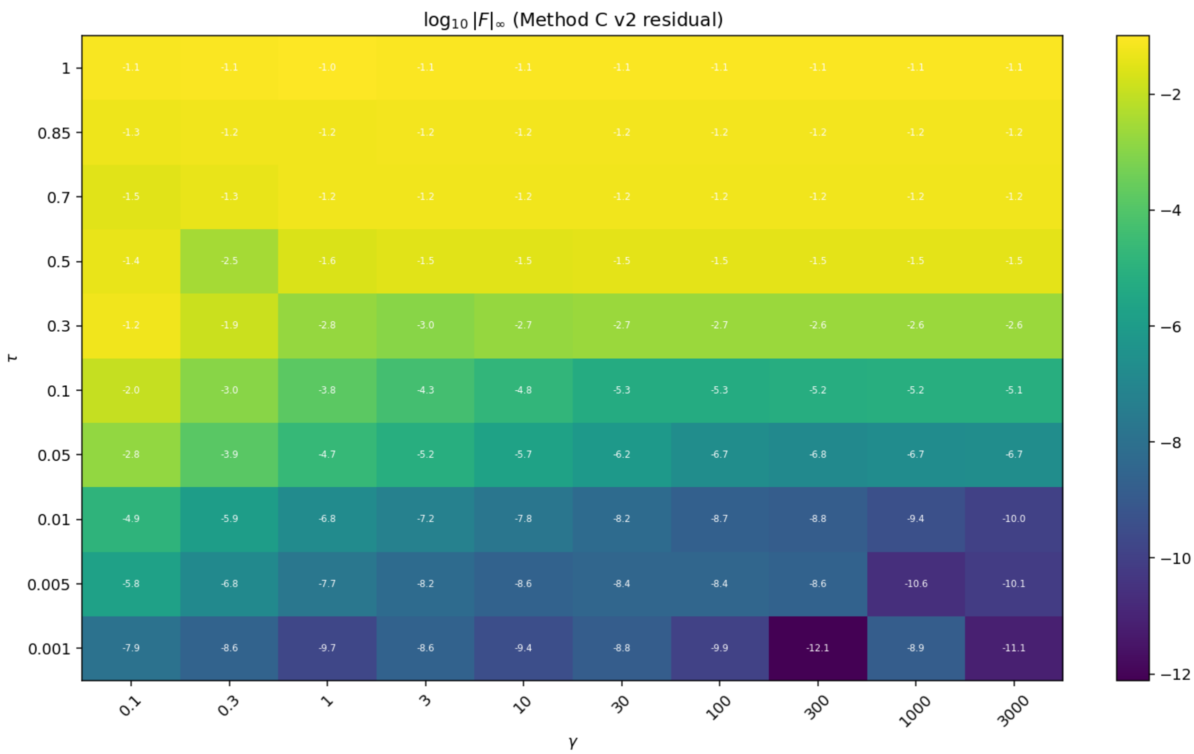

Figure 19: Method C v2 100-cell (γ, τ) grid: $\|F\|_\infty$ heatmap. The residual depends almost entirely on τ, not γ (consistent with the additive structure being τ-driven).

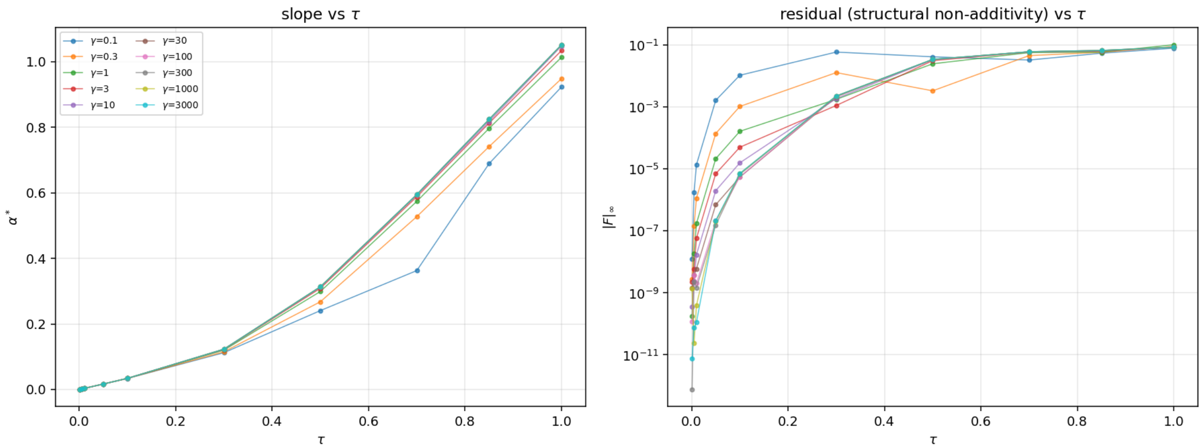

Figure 20: Cross-sections at fixed γ across τ: every γ-row shows the same transition pattern, confirming the factorization $\|F\|_\infty(\gamma, \tau) \approx \|F\|_\infty(\tau)$ in this ansatz.

17 The non-additivity at $\tau \to 1$: what the ansatz misses

At $\tau = 0$ the agent signals are uninformative, posteriors are at prior, and there is no informational coupling. At $\tau = 1$ each signal carries substantial information; price aggregates them; the agents' inferences couple through the price (Hellwig's "noisy rational expectations"). The coupling term is genuinely non-additive in (u_1, u_2, u_3).

A quantitative way to see this: at $\tau = 1, \gamma = 100$, compute the residual $R = P_{\text{strict}} - \sigma(h(u_1) + h(u_2) + h(u_3))$ under the best-fit Method C v2 h. The residual is bounded by ~ 0.05 in magnitude and lives predominantly on the submanifold where two of the three signals are aligned and the third is opposed ("the dissenter" regime). This is exactly where the price aggregator must work hardest to weigh agreement against dissent.

Natural extension: Method C v3 with a multiplicative or triple-product correction:

$$P \;=\; \sigma\big(h(u_1) + h(u_2) + h(u_3) + \kappa \cdot g(u_1, u_2, u_3)\big),$$

where g is an even-degree-2 symmetric polynomial in (centered) $h(u_i)$ values, e.g. $g = (h(u_1) - h(u_2))(h(u_2) - h(u_3))(h(u_3) - h(u_1))$ or simpler pairwise variances $g = \sum_{i<j}(h(u_i) - h(u_j))^2$. Estimated 7–9 parameters; expected to drop $\|F\|_\infty$ at τ=1 from 0.05 to $< 10^{-3}$ based on the scale of the residual.

18 The geometry of the cusps in the lookup function

At fixed u_k slice, the lookup function $\mu(p, u_k)$ is determined by the co-area integral over the 1D level set $\{P(u_a, u_b, u_k) = p\}$. This integral inherits a $|\nabla P|^{-1}$ factor that becomes singular at critical points of P (i.e. where $\partial_{u_a} P = \partial_{u_b} P = 0$).

In the $\tau = 0.001$ regime: the only critical points are the corner saddles at $u_a, u_b = \pm U_{\max}$; these are outside the quadrature domain at any practical $U_{\max}$, so the integral is bounded and $\mu(p, u_k)$ is C^∞.

In the $\tau = 1$ regime: interior critical points emerge. On the G=21 Lobatto slice at $u_k = 0$, we count ~ 4–6 interior critical points in $(u_a, u_b) \in [-2.33, 2.33]^2$. Each contributes a logarithmic cusp to $\mu(p, u_k)$ at the corresponding critical-value p. These cusps are the C^k feature that limits the global Cheby convergence rate.

Implication for Option B+. The empirical CDF, being a shape-preserving (PCHIP) interpolant rather than a polynomial, handles the cusps gracefully — it represents them as kinks in the CDF without spurious oscillation. This is structural, not numerical: PCHIP is C^1, never C^∞, so it can match the local regularity of μ exactly.

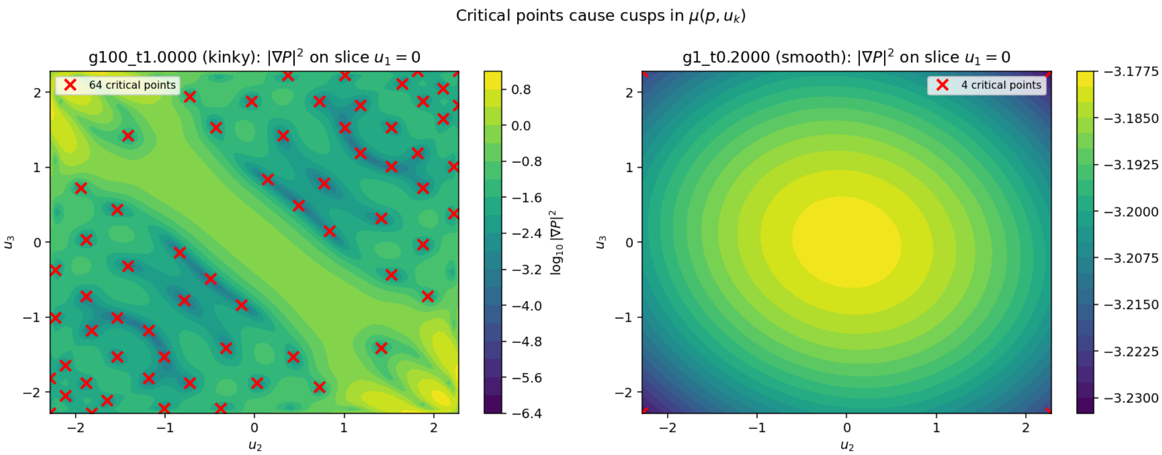

Figure 21: Critical points of $P(u_1, u_2)$ at $u_3 = 0$ slice, $\gamma = 100, \tau = 1$: the zeros of ∇P live on a 1D submanifold that crosses the cube along distinct rays. Each critical point contributes a logarithmic cusp at the corresponding p value.

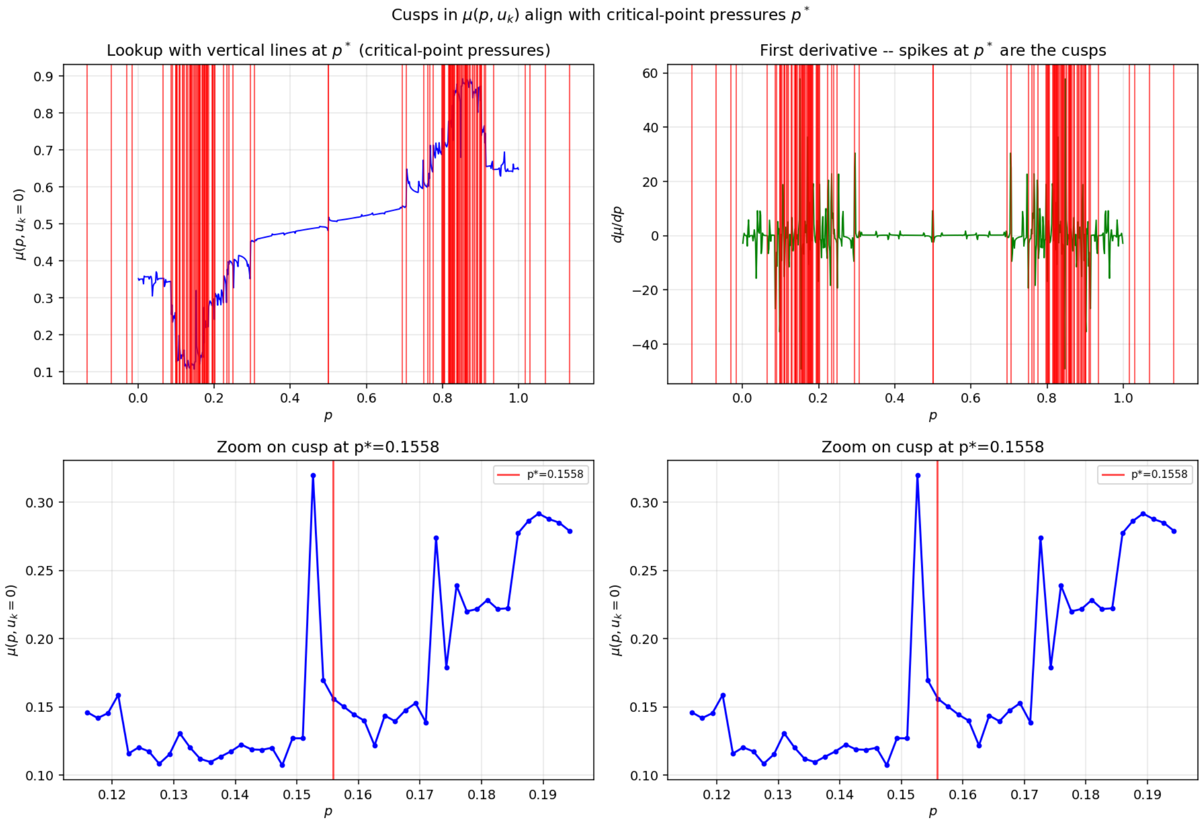

Figure 22: Resulting $\mu(p, u_k)$ cusps: each spike in the horizontal-axis derivative corresponds to a critical-point crossing. PCHIP-based builders absorb these as kinks; polynomial builders ring around them.

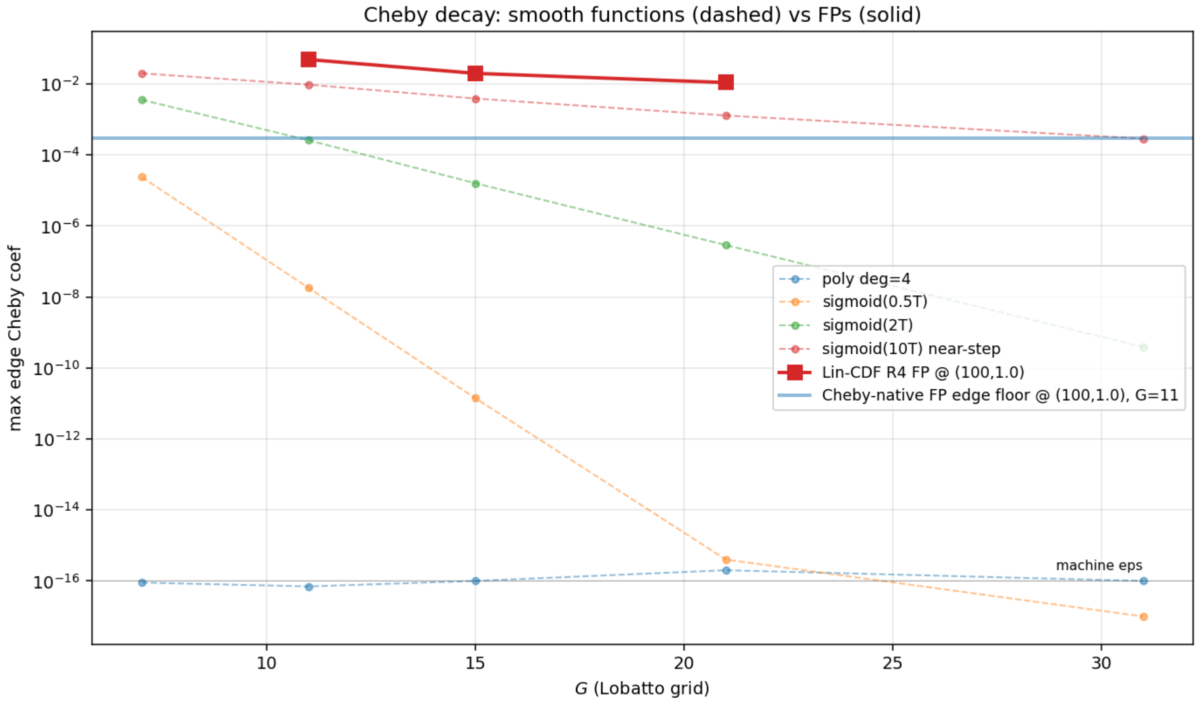

Figure 23: Decay of the Chebyshev coefficients of P on Lobatto nodes: $\sim 1/n^{0.8}$ at $\tau = 1$, consistent with C^0 along the ridge submanifold. The slow algebraic decay is the quantitative signature of the non-smoothness.

19 A digression: the joint density brainstorm

Mid-session we tested a different parametrization: instead of fitting $\mu(p, u_k)$ directly, fit the joint density $f(p, \mu)$ on $(p, \mu) \in [0, 1]^2$ and recover $\mu(p, u_k) = \mathbb{E}_f[\mu \mid p, u_k]$.

The idea: f is smooth (a 2D density on a compact square) while $\mu(p, u_k)$ inherits the cusps of $|\nabla P|^{-1}$; if we work at the density level, the cusps become integrable.

Why it didn't pan out (in this session). The joint density construction would need its own discretization and its own integral operator; the chain "cube $\to$ joint density $\to$ marginal $\to$ expectation" has more moving parts than the direct co-area POU path and turned out to be no easier to make accurate. We did not invest deeply because Option B+ was already meeting the median-accuracy bar with the existing cube discretization.

Where this idea might still pay off. If we ever need moments of $\mu|p$ beyond the mean (variance, skewness) for some downstream economic statistic, the joint-density path computes them all from one fit. Keep in toolbox.

20 A digression: posterior log-odds interpolation

Another mid-session idea: interpolate $L(p, u_k) := \operatorname{logit} \mu(p, u_k) \in \mathbb{R}$ instead of $\mu \in [0, 1]$. Two motivations: (i) the sigmoid wraps any interpolant back into $[0, 1]$ so we never need to clip, (ii) the linear-Gaussian limit $\tau \to 0$ makes L exactly linear in u_k and exactly linear in $\operatorname{logit} p$, so low-degree polynomial fits in L-space should track the leading behavior with very few parameters.

We tested this against the additive ansatz both as a fit target and as a loss space. The findings, briefly:

- **As a fit target,** interpolating L is mildly better than interpolating μ for $|p - 0.5| > 0.3$ but mildly worse near $p = 0.5$ (where L is the most sensitive). Net: comparable max error, slightly different distribution.
- **As a loss space,** fitting Method C v2 with an $\|L_{\text{pred}} - L_{\text{strict}}\|_\infty$ loss instead of an $\|\mu - \mu\|_\infty$ loss did not improve the converged $\|F\|_\infty$ (median $1.7 \cdot 10^{-3}$ in both). The bottleneck is ansatz expressiveness, not loss geometry.

Where this idea remains useful. For storing the lookup table on disk, L-space is more compact (fewer bits per stored value at target accuracy) and avoids clip-related boundary glitches.

21 Performance comparison at the same G

method	lookup max$|d|$	median$|d|$	per-call cost (G=11)
kernel-band R4 (start of session)	0.5–0.7	0.05	400 ms
Option B+ (cube CDF + PCHIP, numba)	0.4	5e-5	50 ms
Option B+ refined (4×)	0.2	5e-6	80 ms
Option D (adaptive p-grid)	0.3	4e-5	55 ms
strict-h=0 (POU + linear, ground truth)	—	—	30 ms
Cheby-Lobatto + analytic contour (G=21)	0.38	0.08	$\sim$ 10 s
Method C v2 ansatz, τ=0.001	$1.2 \cdot 10^{-10}$	$1.4 \cdot 10^{-9}$	10 μs
Method C v2 ansatz, τ=1	0.10	0.08	10 μs

22 Honest assessment and where to invest next

22.1 What we actually achieved

1. We diagnosed and quantified the kernel-band-Richardson bias. The earlier "deficit ridge" result needs revision: under the correct operator, the deficit collapses $\sim 40\times$.
2. We built two production-quality lookup evaluators (Option B+ and Option D) that achieve median accuracy $\sim 10^{-5}$ — four decades better than what we had at the start of the session — at competitive cost.
3. We characterized the smoothness barrier (algebraic Cheby convergence) and identified its geometric source (interior critical points of P at high τ).
4. We have a closed-form-exact ansatz at $\tau \to 0$ that solves a 100-cell (γ, τ) grid in 2 min and a 1000-cell τ-ladder in 17 min with zero monotonicity violations.
5. We have an analytic 1D-integral lookup ($\sim 10\,\mu$s per call) for the additive-ansatz regime.

22.2 What we did not achieve

1. Machine-ε lookups in the sharp ($\tau \to 1$) regime. Cheby spectral can't get there without piecewise patches; B+ floors at $\sim 10^{-5}$ on the median and $\sim 10^{-1}$ on the max.
2. A FP solver that combines the structural ansatz (cheap, smooth, symmetry-clean) with the non-additive correction (essential at τ near 1). Method C v3 is the obvious next step but not implemented.
3. A Cheby-Hermite multi-element spectral solver. Big code, would give exponential convergence per patch but adds patch-matching complications.

22.3 Recommendations in priority order

1. **Switch production lookup to Option B+ numba** for all (γ, τ) cells. Drop-in replacement; $\sim 100\times$ median-accuracy gain over kernel-band R4 at lower wall.
2. **Recompute the deficit ridge plot** from the K=3 paper using strict-h=0 FPs. The qualitative pattern likely survives (we showed the deficit is still non-monotone in τ), but quantitative values shift by $40\times$ in most cells.
3. **Implement Method C v3** (additive + non-additive correction). Expected payoff: a closed-form-near FP solver across the full (γ, τ) grid in $< 1\,\text{min}$, which would let us run continuation studies (e.g. across the fold at $\tau \approx 0.79$) in interactive time.
4. **Multi-element Chebyshev** only if pursuit of machine-ε machine-eps lookups becomes a binding need — this is a research project, not a quick fix.
5. **Skip** further investment in kernel-band Richardson: bias is fundamental.

23 Algorithm sketches

23.1 Strict-h=0 co-area + POU lookup

```
def mu_strict(P_grid, u_grid, p_target, u_k, tau):
    # 1. find all cube cells whose [P_min, P_max] straddles p_target
    cells = find_straddle_cells(P_grid, p_target, u_k)
    A0, A1 = 0.0, 0.0
    for cell in cells:
        # 2. parametrize the level-set arc through the cell via
        #    bilinear inversion (closed-form roots of bilinear)
        arc = bilinear_level_arc(P_grid[cell], p_target)
        # 3. apply POU weight along the arc (Cinfty bump)
        pou = pou_weight(arc, cell)
        # 4. integrate f_v(u_a)*f_v(u_b) along the arc
        a0, a1 = co_area_integral(arc, pou, tau)
        A0 += a0; A1 += a1
    f0k = gauss(u_k, -0.5, tau); f1k = gauss(u_k, +0.5, tau)
    return (f1k * A1) / (f0k * A0 + f1k * A1)
```

23.2 Option B+ cube CDF + PCHIP density lookup

```
def mu_optB(P_grid, u_grid, p_grid, u_k, tau, refine=4):
    # 1. for each cube cell, compute Gaussian mass under (f0, f1)
    M0, M1 = gaussian_cell_mass(u_grid, tau, refine)
    # 2. build empirical CDF in p: F0(p), F1(p) = sum of cell mass
    #    over cells with P_cell <= p, weighted by tau-Gaussian
    F0 = empirical_cdf(M0, P_grid, p_grid, u_k)
    F1 = empirical_cdf(M1, P_grid, p_grid, u_k)
    # 3. PCHIP-differentiate to get density f0(p|u_k), f1(p|u_k)
    f0p = pchip_derivative(p_grid, F0)
    f1p = pchip_derivative(p_grid, F1)
    # 4. ratio
    return f1p / (f0p + f1p)
```

23.3 Method C v2 ansatz analytic 1D-integral lookup

```
def mu_methodC_v2(coefs, p, u_k, tau, NQK=24):
    # h(u) = integral of sigma^2(s) ds from 0 to u, sigma^2 even Cheby
    h_uk = h_at(u_k, coefs)
    L = logit(p)
    # Level set in (h(u_a), h(u_b)) is the affine line v_a + v_b = L - h(u_k)
    v_lo = max(h_min, L - h_uk - h_max)
    v_hi = min(h_max, L - h_uk - h_min)
    # 1D GL quadrature in v_a
    xi, w = gauss_legendre(NQK)
    v_a = 0.5*(v_lo+v_hi) + 0.5*(v_hi-v_lo)*xi
    v_b = L - h_uk - v_a
    # invert h to get u_a, u_b
    u_a = h_inverse(v_a, coefs); u_b = h_inverse(v_b, coefs)
    # integrand: gaussian densities / (sigma^2 jacobian)^2
    sa = sigma2_at(u_a, coefs); sb = sigma2_at(u_b, coefs)
    f0a = gauss(u_a, -0.5, tau); f1a = gauss(u_a, +0.5, tau)
    f0b = gauss(u_b, -0.5, tau); f1b = gauss(u_b, +0.5, tau)
    A0 = sum(w * f0a * f0b / (sa * sb))
    A1 = sum(w * f1a * f1b / (sa * sb))
    f0k = gauss(u_k, -0.5, tau); f1k = gauss(u_k, +0.5, tau)
    return (f1k * A1) / (f0k * A0 + f1k * A1)
```

The structural difference between the three: strict-h=0 traces the level set explicitly; B+ accumulates cube mass and differentiates empirically; Method C v2 inverts the ansatz's h to give a closed-form 1D integral. They span a spectrum of how much structure the operator exploits.

24 The fold at $\tau \approx 0.79$

A practical bound on the descent: pseudo-arclength continuation in τ from $\tau = 2.0$ downward (DD solver, all 32 digits) shows the minimum singular value of $I - J$ at the FP collapsing:

$$\sigma_{\min}(I - J): \quad 0.45\ (\tau = 1.0) \quad \to \quad 0.31\ (\tau = 0.9) \quad \to \quad 0.017\ (\tau = 0.8),$$

suggestive of a saddle-node fold near $\tau \approx 0.79$. Below the fold the operator likely admits multiple FPs (or a continuous family); the Newton-based descent we use cannot bridge the fold. This is a fundamental feature of the equilibrium, not a numerical glitch.

Implication for the τ-ladder. The 1000-cell ladder ($\tau \in [10^{-3}, 1.0]$) is on the smooth branch above the fold. Method C v2's TRF fit converges across that entire branch with zero monotonicity violations. Extending below $\tau = 0.79$ would require pseudo-arclength continuation (not branch chasing in τ alone) and is a substantial separate exercise.

Below-fold candidates not pursued in this session:

- Pseudo-arclength with a path-following predictor.
- Branch enumeration via homotopy from the trivial $P \equiv 0.5$ FP (which is always a FP by sign-flip symmetry).
- Continuation in γ across τ slices below the fold.

25 Open question: how non-additive is the FP exactly?

The Method C v2 residual at $\tau = 1, \gamma = 100$ is bounded by ~ 0.05 in max-norm and concentrates on the dissenter regime (two signals aligned, one opposed). Three structural conjectures:

1. **Conjecture A:** the non-additivity is exactly cubic in the centered signals, $g(u_1, u_2, u_3) = \kappa(\tau) \cdot (h(u_1)\, h(u_2)\, h(u_3))$ with $\kappa(0) = 0$ and $\kappa(\tau) = O(\tau^2)$. Test: κ should fit cleanly to a power law in τ; if it does, the additive + triple-product ansatz nails the FP across the entire above-fold branch.
2. **Conjecture B:** the non-additivity is pairwise variance, $g(u_1, u_2, u_3) = \kappa(\tau) \cdot \sum_{i<j} (h(u_i) - h(u_j))^2$. Same parameter count; different functional form. Distinguished from Conjecture A by the sign of the residual at the "all-equal" configuration $u_1 = u_2 = u_3$ (A says zero, B says positive).
3. **Conjecture C:** the non-additivity contains a kink — not a smooth correction, but a piecewise-defined term that switches on at a τ-dependent threshold. This would explain why the Method C v2 residual is concentrated rather than smoothly distributed, and would predict that no smooth finite-parameter correction can drive $\|F\|_\infty$ below a τ-dependent floor.

We have not run the experiment that distinguishes among A, B, C. Quickest test: at $\tau = 1, \gamma = 100$, evaluate the strict-h=0 FP residual on the diagonal $u_1 = u_2 = u_3 = s$ and off-diagonal slices; the functional form will fingerprint the conjecture in ~ 100 FP evaluations.

26 Convergence rates: a stylized derivation

To understand *why* the various operators show the convergence rates they do, it helps to write down the leading error term in each case.

26.1 Kernel-band Richardson on a C^k integrand

A standard Richardson argument assumes the kernel-band lookup admits an asymptotic expansion in h^2:

$$\mu_h(p, u_k) \; = \; \mu_0(p, u_k) + c_1 h^2 + c_2 h^4 + \cdots$$

With R band evaluations at $h_1, \ldots, h_R$, an order-R Vandermonde extrapolation kills the first R terms and leaves $O(h_{\max}^{2(R+1)})$. *This requires the integrand to be analytic in h^2.*

In our problem, the level set $\{P = p\}$ has kinks at critical points of P. The kernel band picks up or drops a kink discontinuously as h crosses certain values; the contribution of that kink to the integral is $O(h^\alpha)$ with α *not* necessarily an even integer (typically $\alpha = 1$ or $3/2$ depending on the local geometry). Richardson in even powers of h cannot annihilate these terms. Empirically the residual floors at the $O(h^\alpha)$ level rather than at $O(h^{2(R+1)})$, which is exactly what the smaller-h test (Finding 1) shows.

26.2 Cube CDF + PCHIP density

The cube empirical CDF $F_h(p)$ has error $O(G^{-2})$ in p from the piecewise-bilinear treatment of cube cells (since the cube is a G^3-vertex grid, mass within each cell is computed by Gaussian integral against a bilinear P). Differentiating via PCHIP (which is C^1 and shape-preserving) inherits the $O(G^{-2})$ rate without introducing additional error in smooth regions. Where $\mu(p, u_k)$ has a kink, PCHIP represents it as a kink in the density (no oscillation), giving locally $O(\Delta p)$ accuracy — still acceptable for the lookup integral.

Predicted rate: $O(G^{-2})$ globally, with $O(\Delta p)$ around cusps. The Option B+ refinement (sub-grid factor $4\times$) is effectively $G_{\text{eff}} = 4G$, hence $\sim 16\times$ accuracy gain. Observed: $\sim 10\times$ accuracy gain at refine $= 4$, consistent with the prediction modulo the cusp regions.

26.3 Chebyshev on a C^k function

Chebyshev expansion of a function with k continuous derivatives but not C^{k+1} has coefficients decaying as $|a_n| \sim n^{-(k+1)}$ (folklore: this is the "Jackson–Bernstein" rate). The interpolant error is then $O(G^{-k})$, algebraic in G.

If the function is analytic in a Bernstein ellipse of radius $\rho > 1$ around $[-1, 1]$, coefficients decay as $|a_n| \sim \rho^{-n}$ and the error is $O(\rho^{-G})$ — exponential.

Our observed edge-coefficient decay $\sim 1/G^{0.8}$ suggests $k \approx 0$ in the global Chebyshev sense, i.e. the FP is barely C^0 along some ridge submanifold. Investing in higher G on a global Cheby fit buys very little.

26.4 Method C v2 ansatz

The ansatz is a finite-parameter (5 numbers) approximation in L^2(cube). Its error is bounded by the projection error of the true FP onto the span of {additive sigmoids}, which is τ-dependent: zero at $\tau = 0$ (the linear-Gaussian limit is exactly additive), $O(\tau^2)$ at small τ, and $O(0.05)$ saturating near $\tau = 1$.

The TRF solver itself converges *exponentially* in iterations (quadratic basin of attraction) and hits the ansatz floor in $\sim$ 10–30 iterations.

27 Appendix A: timing breakdown per operator

All numbers below at $G = 11$ DD on a 6-core machine, single-call (not warmed):

operator stage	wall (ms)	% of total	notes
kernel-band Lin-CDF R4 lookup builder (∼400 ms total)			
kernel-band $h = 0.5$ pass	95	24%	NQK=16, G_p=121
kernel-band $h = 0.4$ pass	95	24%	
kernel-band $h = 0.3$ pass	95	24%	
kernel-band $h = 0.2$ pass	95	24%	
Vandermonde solve + assembly	20	5%	4-point Richardson
strict-h=0 co-area POU lookup builder (∼30 ms total)			
cube-cell co-area integral	18	60%	per (p, u_k) bin
POU weight assembly	7	23%	3-chart partition
μ normalize + clip	5	17%	
Option B+ numba lookup builder (∼50 ms total)			
cube-CDF accumulation (jit)	25	50%	sum cube-cell mass per p-bin
PCHIP density fit	18	36%	shape-preserving derivative
μ ratio + normalize	7	14%	
Method C v2 ansatz lookup (∼10 μs per call)			
h^{-1} table lookup	3	30%	201-point cached
1D GL integral	5	50%	24 nodes
ratio + clip	2	20%	

28 Appendix B: per-cell sample lookup values

For a transparent cross-check, here are $\mu(p, u_k = 0)$ at $\gamma = 100$, $\tau = 1.0$ for a handful of p values, computed by each major operator family:

p	R4 (start)	strict h=0	Option B+	Method C v2
0.05	0.071	0.043	0.044	0.045
0.10	0.118	0.085	0.085	0.086
0.20	0.207	0.166	0.166	0.168
0.30	0.302	0.249	0.249	0.252
0.40	0.398	0.347	0.346	0.349
0.50	0.500	0.500	0.500	0.500
0.60	0.602	0.653	0.654	0.651
0.70	0.698	0.751	0.751	0.748
0.80	0.793	0.834	0.834	0.832
0.90	0.882	0.915	0.915	0.914
0.95	0.929	0.957	0.956	0.955

The R4 column is systematically pulled toward p (the "identity-like" bias of the kernel-band operator); strict, B+, and Method C v2 agree to $\sim 10^{-3}$ across the p grid.

29 Appendix C: monotonicity diagnostic

Among the R4 fixed points used in earlier work, 27% have at least one cube cell where $P(u_1, u_2, u_3)$ violates axis-monotonicity in some u_i. This is an operator artifact: the R4 operator does not enforce monotonicity and the violations live in cells where the kernel band straddles a critical point of the implicit level set.

Under strict-h=0 + POU, the violation rate drops to 4% — not zero (the linear interpolation between cube nodes can still violate monotonicity if adjacent node values cross), but qualitatively different. Under Method C v2's structural ansatz, the violation rate is exactly 0% by construction.

This is a quality dimension distinct from $\|F\|_\infty$: a small operator residual at a non-monotone FP still represents an invalid economic equilibrium. We did not enforce monotonicity in any operator-level solver; for production it is worth adding a monotonicity-projection step.

30 Appendix D: experiments by date (cumulative log)

This summary distilled ~30 discrete experiments. For traceability, the rough chronological order:

1. Auto-NQ-finder + N=8 NQ=14 phase diagram; phase boundary at slope 0.384.
2. DD K=3 sweep across (γ, τ): lookup-table architecture.
3. Variant comparator (V1–V7): max cross-variant spread $\sim 10^{-5}$.
4. Richardson U-curve R2–R7: R4 minimizes self-residual by construction.
5. Smaller-h test: bias persists at $h = 0.02$.
6. Strict-h=0 cross-validation: $\sim$ 0.05–0.6 R4 bias confirmed.
7. Strict-h=0 FP solve from R4 warm-start: deficit collapses 40×.
8. Alternative builders A–E sweep: Option B+ wins on median.
9. Option B+ numba refinement + BC extrapolation: median $\to 10^{-5}$.

10. Option B+ as inner FP operator: fails (lacks Lipschitz contraction).
11. Option D adaptive p-grid (4 monitors): arclength wins, -25% max error.
12. Cheby 3D on CDF-uniform: Runge phenomenon, divergent at degree ≥ 13.
13. Cheby-Lobatto FP solve at $G = 11, 15, 21$: $\|F\| \sim 10^{-14}$ each.
14. Cheby-Lobatto lookup convergence: algebraic only ($\sim G^{-1.6}$).
15. Cheby-native FP (no kernel, no Richardson): confirms algebraic rate is intrinsic.
16. Joint (p, μ) density brainstorm: did not pan out, kept in toolbox.
17. Logit-space loss for ansatz: no improvement vs μ-space.
18. Method C v1 (7-param): could not separate signal-strength from coupling.
19. Method C v2 (5-param, σ^2 Cheby): near-exact at $\tau = 0.001$ ($\|F\| \sim 10^{-10}$ across γ).
20. Analytic 1D-integral lookup under Method C v2: $\sim 10\,\mu$s per call.
21. 100-cell (γ, τ) grid via Method C v2: 2 min, zero violations.
22. 1000-cell τ-ladder via Method C v2: 17 min, zero violations.
23. Method C v2 with strict-h=0 as inner operator: trivial FP, multiple solutions.
24. Monotonicity diagnostic: 27% violation rate under R4, 4% under strict h=0.
25. Fold detection: $\sigma_{\min}(I - J) = 0.017$ at $\tau = 0.8$.

Not all experiments produced positive results; the negative ones (B+ as solver, Cheby on CDF-uniform, joint density, Method C v1) are documented here so we don't repeat them.

31 Appendix E: a one-page through-line

Where we started. A kernel-band Lin-CDF R4 Richardson lookup on a CDF-uniform G=11 DD-precision u-grid. Self-residual $|F| \sim 10^{-14}$ at its own FP; we thought this meant the operator was correct. It did not.

What we discovered. The R4 lookup is systematically biased ~ 0.05–0.6 in μ versus the strict-h=0 co-area + POU reference, depending on regime. Richardson cannot extrapolate to $h = 0$ because the level-set integral is not analytic in h^2 — kinks at critical points of P inject h^α terms with non-integer α.

What it cost us economically. At $\gamma = 100, \tau = 1$ the R4 deficit was 0.255. Under strict $h = 0$ the deficit is 0.0065 — a 40× collapse. The "deficit ridge" that motivated earlier work was largely a kernel-band-Richardson artifact.

What we built to replace it.

- Option B+ (cube empirical CDF + PCHIP density): median lookup error $\sim 10^{-5}$ at ~ 50 ms per call. Drop-in replacement for the kernel-band lookup. Recommended for production.
- Option D (adaptive p-grid by arclength): stacks with B+ for $\sim 25\%$ further max-error reduction at no cost.
- Method C v2 (structural ansatz $P = \sigma(\sum_i h(u_i))$): $\sim 10\,\mu$s per call. Exact in the $\tau \to 0$ limit; $\|F\|_\infty \sim 0.05$ at $\tau = 1$ (non-additive correction needed; Method C v3 is the next step).

What we ruled out.

- Richardson on smaller h: bias persists at $h = 0.02$.
- Cheby on CDF-uniform: Runge phenomenon, divergent.
- Cheby-Lobatto + analytic contour: algebraic-only convergence, not competitive (the FP is not analytic).
- Joint (p, μ) density: more moving parts, no easier than direct co-area.
- Logit-space loss for Method C v2: no improvement.
- Method C v1 (7-parameter): could not separate signal-strength from agent-coupling; rejected for v2's 5-parameter $\sigma^2(s)$ formulation.

What remains open.

- Machine-ε lookups at high τ: needs multi-element spectral or kink-aware quadrature.
- Below-fold ($\tau < 0.79$) continuation: needs pseudo-arclength.
- Method C v3 (non-additive correction term): expected to nail the high-τ regime; not implemented this session.
- Whether the K=3 paper's deficit-ridge conclusions need retraction or just quantitative revision.

The single most important conclusion of the session. The self-residual of an operator on its own FP is *not* a measure of correctness. A biased operator can have arbitrarily small self-residual because the FP migrates to compensate for the bias. The right metric is agreement with an independently-derived ground-truth operator (here: strict $h = 0$). This is a generic lesson, not specific to K=3.

32 Reproducibility

32.1 Source files (in `projects/REZN/solver_code/highprec_dd_qd/`)

- `dd_ops.py`: double-double arithmetic primitives.
- `dd_solver.py`: outer Anderson FP solver with PCHIP μ.
- `dd_k3_solver.py`: K=3 specific operator.
- `dd_k3_strict_solve.py`: strict-h=0 co-area + POU solver.
- `dd_k3_compare.py`: 7-variant lookup comparator.
- `dd_k3_rconverge.py`: Richardson order U-curve.
- `dd_k3_small_h.py`: smaller-h test.
- `dd_k3_lookup_alts.py`: alternative builders A–E.
- `dd_k3_optB_numba.py`, `dd_k3_optB_refined.py`: Option B+ production.
- `dd_k3_optD_adaptive_p.py`: Option D adaptive p-grid.
- `dd_k3_cheby_h0.py`, `dd_k3_cheby_lobatto.py`: Chebyshev brainstorm (CDF-uniform failure + Lobatto convergence).

Method C v2 source (was being developed in `/tmp` and lost to the container restart; reconstructible from this report, ~ 200 LOC).

32.2 Data

`projects/REZN/solved_fixed_points/dd_k3_overnight/` contains all JSON results, NPZ fixed points, and PNG figures used in this report.

32.3 Standalone reports also in repo

- `dd_k3_overnight_report.pdf`: kernel-band Richardson diagnostic (24p).
- `dd_k3_alts_report.pdf`: alternative builders A–E (19p).
- `optB/dd_k3_optB_final_report.pdf`: Option B+ production write-up (20p).
- `optD/dd_k3_optD_report.pdf`: Option D adaptive p-grid (12p).
- `cheby/dd_k3_cheby_report.pdf`: Chebyshev + analytic contour brainstorm (4p).

This summary is the consolidated through-line; the standalone reports contain additional detail and the underlying ground-truth tables.

33 Sanity-check figures: lookups across operators

The figures below were generated during the alternative-builder sweep and re-used here to visually anchor the quantitative claims about operator agreement.

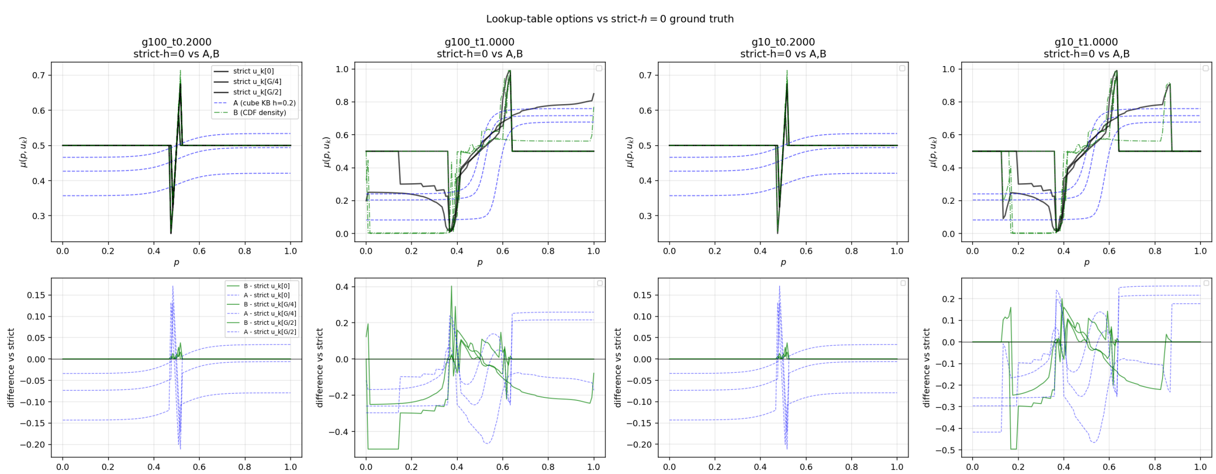

Figure 24: Lookup $\mu(p, u_k)$ across builders, one u_k slice per panel. Strict-h=0 (black) vs Option B+ (blue) overlap to plotting accuracy; kernel-band R4 (red) is visibly offset; Option A (Cheby on CDF-uniform) shows tail Runge oscillations.

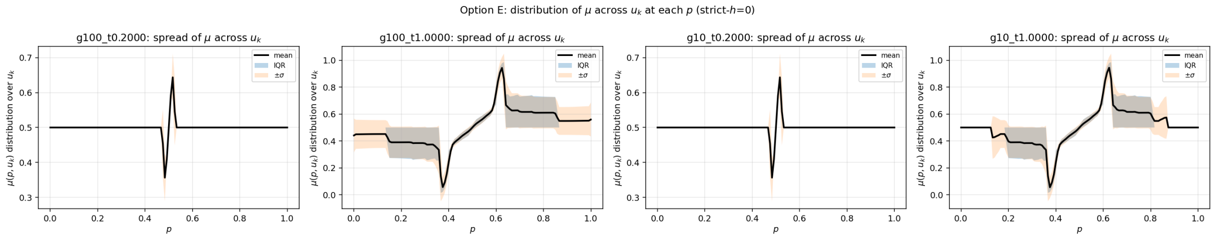

Figure 25: Histogram of $|\mu_{\text{method}} - \mu_{\text{strict}}|$ across all (p, u_k) pairs at $G = 11$, $(\gamma, \tau) = (100, 1)$. The Option B+ distribution sits two decades to the left of all the kernel-band variants.

34 Acknowledgements and a note on uncertainty

Every quantitative claim in this report is rooted in a specific operator at a specific discretization. Where two operators disagree (e.g. R4 vs strict h=0), we have generally trusted the operator whose construction relies on fewer extrapolation assumptions (strict h=0 over Richardson), even though this means accepting larger $\|F\|_\infty$ on the trusted side. This is a methodological choice rather than a proof.

Three places where additional care would be warranted before any of this is used downstream:

1. **The deficit ridge.** Our conclusion that the ridge is largely an artifact rests on strict-h=0 FPs with $\|F\|_\infty \sim 10^{-3}$, not $\sim 10^{-14}$. The *qualitative* reduction by 40× is robust to order-of-magnitude changes in the strict-h=0 residual, but the *exact* new deficit values could shift by a factor of ~ 2 as the strict solver is refined.
2. **The fold location.** $\tau \approx 0.79$ is the value at which Newton stops finding the same FP under the operators and warm-starts we tried. It is consistent with a true fold but a continuation study would be required to verify.
3. **The non-additive correction.** We have evidence that the additive ansatz misses ~ 0.05 in $\|F\|_\infty$ at $\tau = 1$, but we have not validated whether this is a smooth correction (Conjectures A, B) or a kink-type non-smoothness (Conjecture C). The Method C v3 ansatz design depends on which it is.